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(54) **LIBRARY DEVELOPMENT ENSURING LIBRARY COMPATIBILITY AND MINIMIZING DEPENDENCY CONFLICTS**

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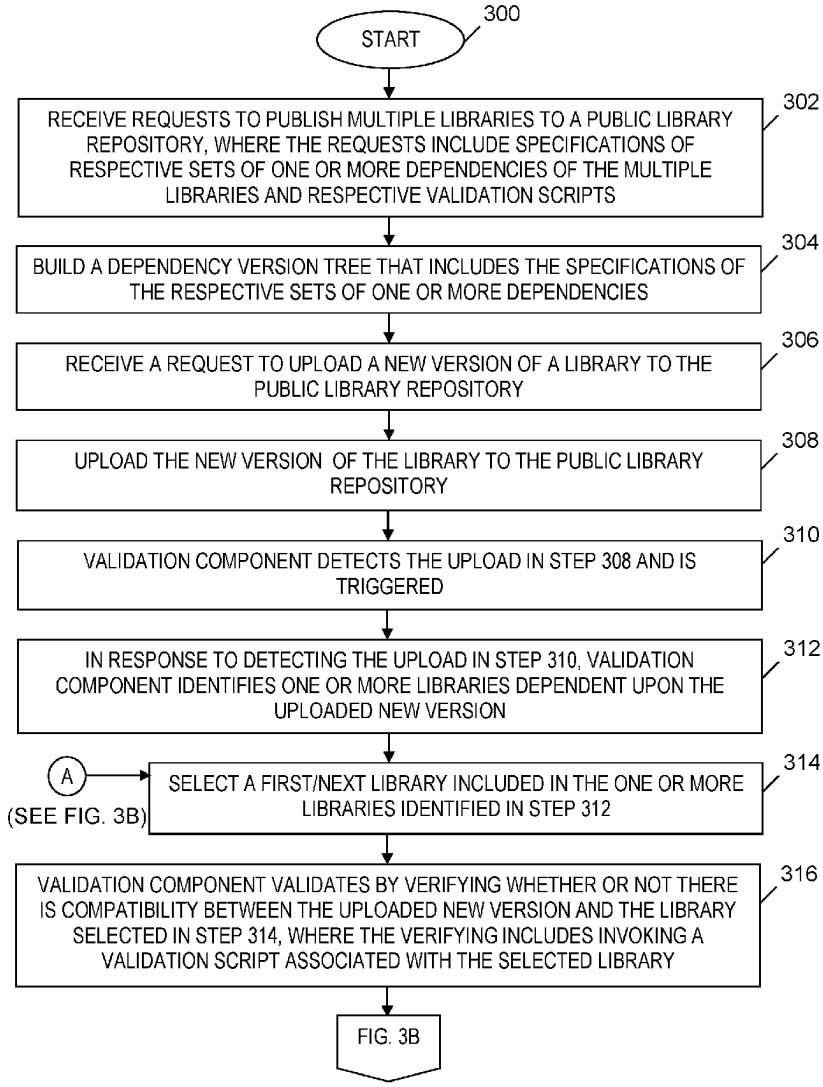
(57) **ABSTRACT**

An approach is provided for managing library dependencies. Requests to publish multiple libraries to a public library repository are received. The requests include specifications of respective sets of one or more dependencies of the libraries and respective validation scripts. A dependency version tree that includes the specifications of the respective sets of one or more dependencies is built. A request to upload a new version of a library to the public library repository is received. A validation component using the dependency version tree and the validation scripts validates a compatibility between the library and one or more libraries dependent upon the library.



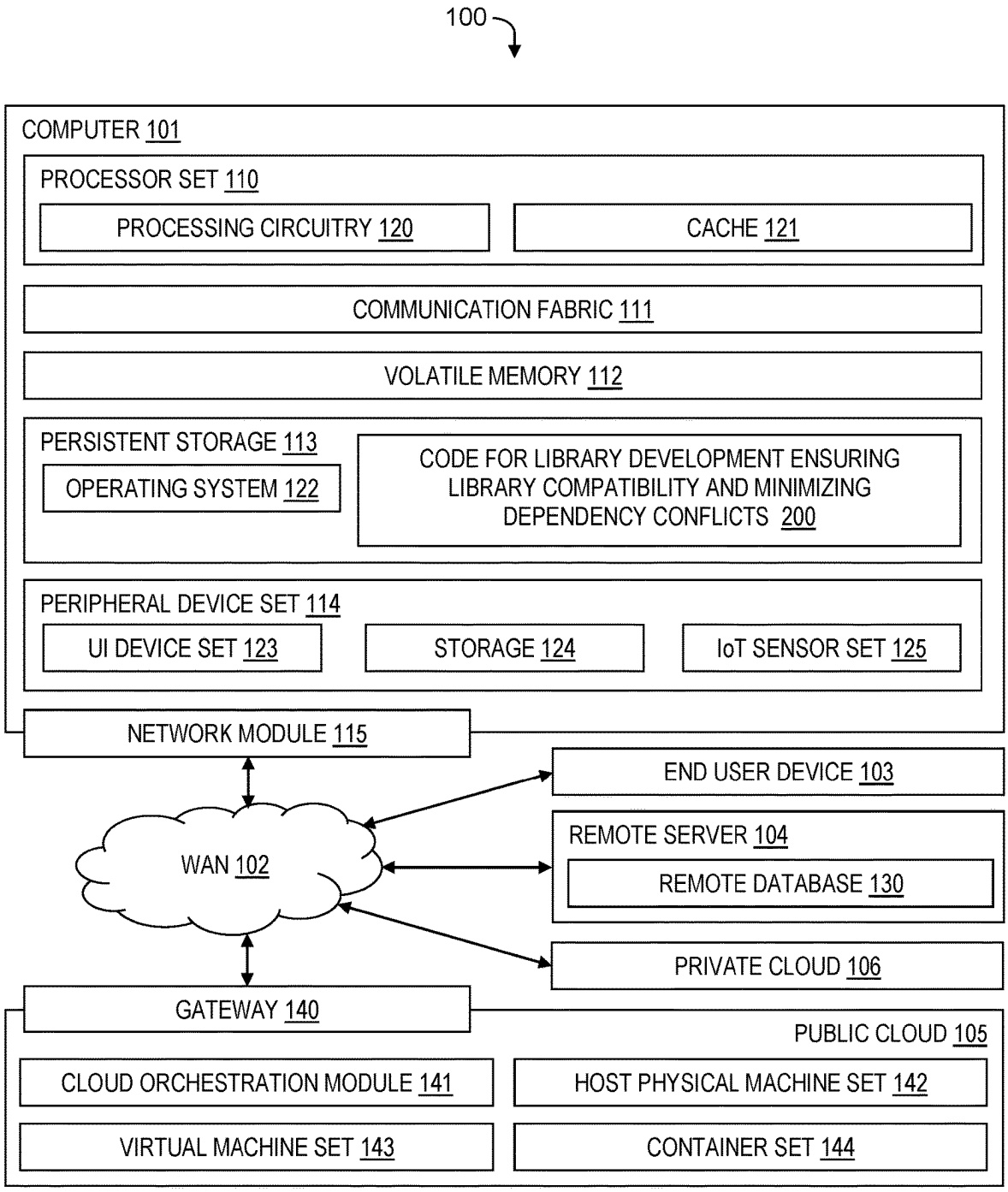

100
COMPUTER 101
PROCESSOR SET 110
PROCESSING CIRCUITRY 120
CACHE 121
COMMUNICATION FABRIC 111
VOLATILE MEMORY 112
PERSISTENT STORAGE 113
OPERATING SYSTEM 122
CODE FOR LIBRARY DEVELOPMENT ENSURING LIBRARY COMPATIBILITY AND MINIMIZING DEPENDENCY CONFLICTS 200
PERIPHERAL DEVICE SET 114
UI DEVICE SET 123
STORAGE 124
IoT SENSOR SET 125
NETWORK MODULE 115
WAN 102
END USER DEVICE 103
REMOTE SERVER 104
REMOTE DATABASE 130
PRIVATE CLOUD 106
GATEWAY 140
PUBLIC CLOUD 105
CLOUD ORCHESTRATION MODULE 141
HOST PHYSICAL MACHINE SET 142
VIRTUAL MACHINE SET 143
CONTAINER SET 144

*FIG. 1*

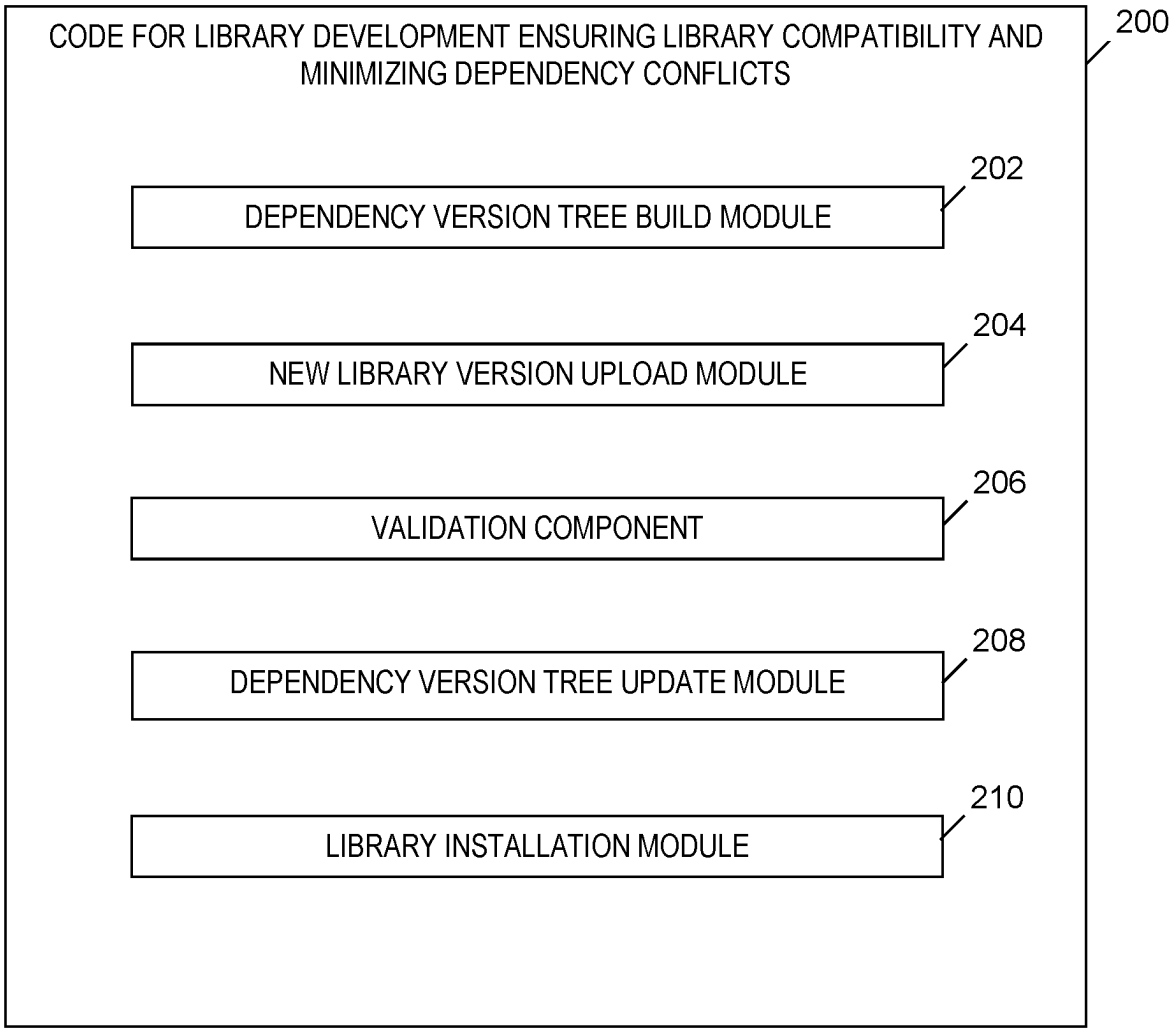

CODE FOR LIBRARY DEVELOPMENT ENSURING LIBRARY COMPATIBILITY AND MINIMIZING DEPENDENCY CONFLICTS
200
DEPENDENCY VERSION TREE BUILD MODULE
202
NEW LIBRARY VERSION UPLOAD MODULE
204
VALIDATION COMPONENT
206
DEPENDENCY VERSION TREE UPDATE MODULE
208
LIBRARY INSTALLATION MODULE
210

*FIG. 2*

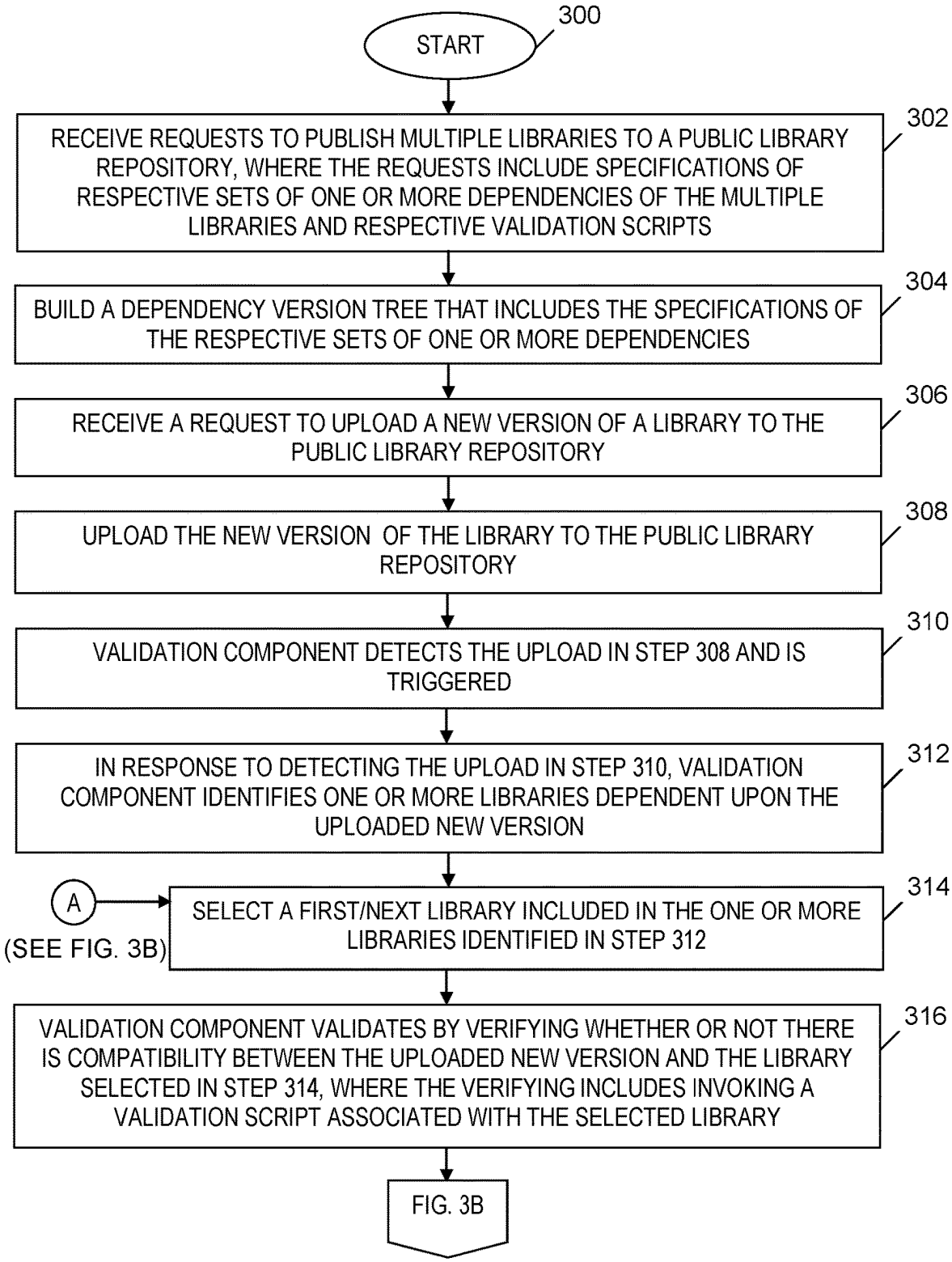

300
START
302
RECEIVE REQUESTS TO PUBLISH MULTIPLE LIBRARIES TO A PUBLIC LIBRARY REPOSITORY, WHERE THE REQUESTS INCLUDE SPECIFICATIONS OF RESPECTIVE SETS OF ONE OR MORE DEPENDENCIES OF THE MULTIPLE LIBRARIES AND RESPECTIVE VALIDATION SCRIPTS
304
BUILD A DEPENDENCY VERSION TREE THAT INCLUDES THE SPECIFICATIONS OF THE RESPECTIVE SETS OF ONE OR MORE DEPENDENCIES
306
RECEIVE A REQUEST TO UPLOAD A NEW VERSION OF A LIBRARY TO THE PUBLIC LIBRARY REPOSITORY
308
UPLOAD THE NEW VERSION OF THE LIBRARY TO THE PUBLIC LIBRARY REPOSITORY
310
VALIDATION COMPONENT DETECTS THE UPLOAD IN STEP 308 AND IS TRIGGERED
312
IN RESPONSE TO DETECTING THE UPLOAD IN STEP 310, VALIDATION COMPONENT IDENTIFIES ONE OR MORE LIBRARIES DEPENDENT UPON THE UPLOADED NEW VERSION
A
(SEE FIG. 3B)
314
SELECT A FIRST/NEXT LIBRARY INCLUDED IN THE ONE OR MORE LIBRARIES IDENTIFIED IN STEP 312
316
VALIDATION COMPONENT VALIDATES BY VERIFYING WHETHER OR NOT THERE IS COMPATIBILITY BETWEEN THE UPLOADED NEW VERSION AND THE LIBRARY SELECTED IN STEP 314, WHERE THE VERIFYING INCLUDES INVOKING A VALIDATION SCRIPT ASSOCIATED WITH THE SELECTED LIBRARY
FIG. 3B

*FIG. 3A*

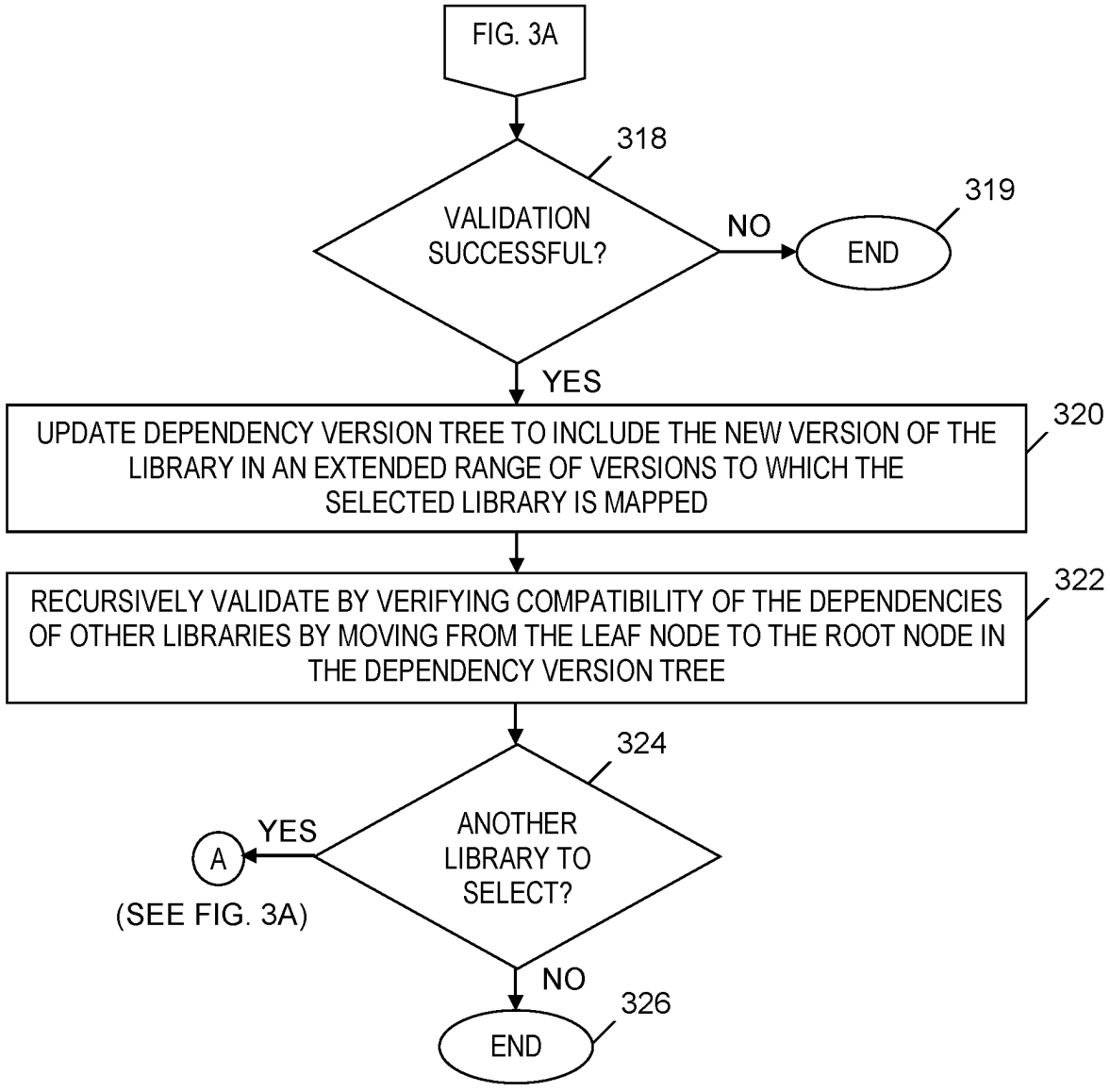


FIG. 3A
318
VALIDATION SUCCESSFUL?
NO
319
END
YES
UPDATE DEPENDENCY VERSION TREE TO INCLUDE THE NEW VERSION OF THE LIBRARY IN AN EXTENDED RANGE OF VERSIONS TO WHICH THE SELECTED LIBRARY IS MAPPED
320
RECURSIVELY VALIDATE BY VERIFYING COMPATIBILITY OF THE DEPENDENCIES OF OTHER LIBRARIES BY MOVING FROM THE LEAF NODE TO THE ROOT NODE IN THE DEPENDENCY VERSION TREE
322
324
ANOTHER LIBRARY TO SELECT?
YES
A
(SEE FIG. 3A)
NO
326
END

*FIG. 3B*

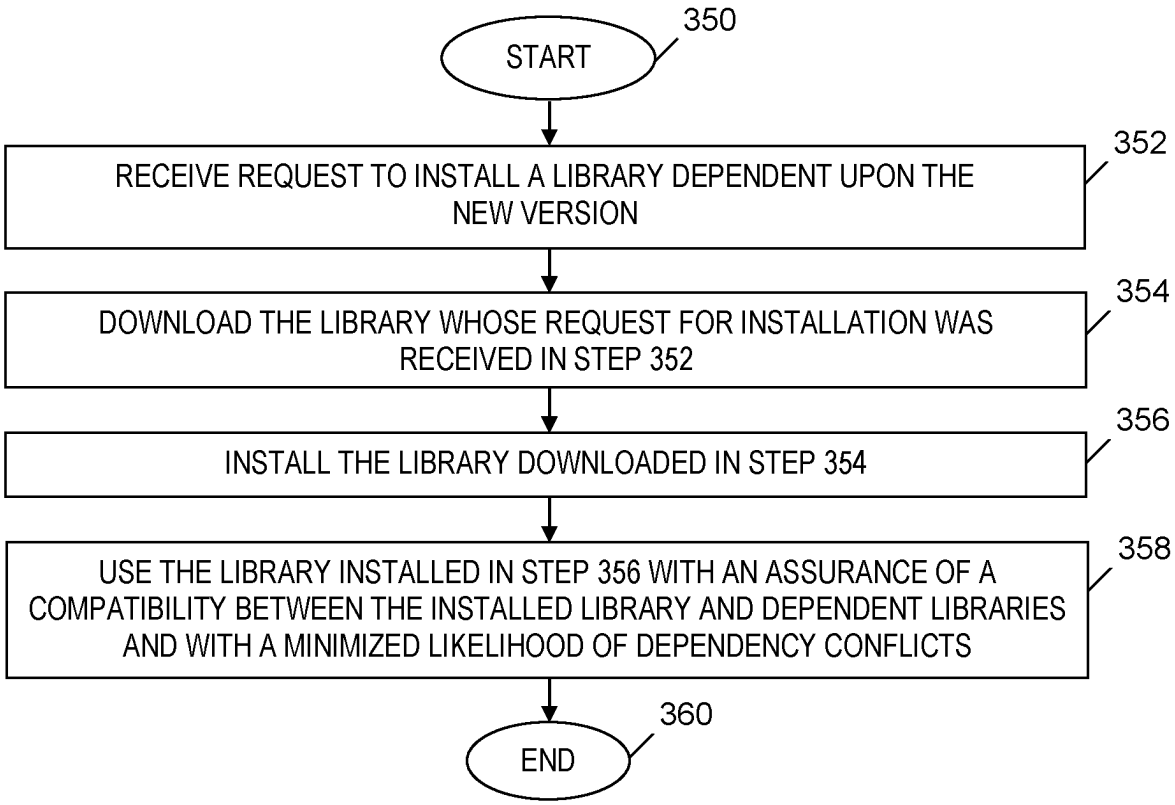

350
START
352
RECEIVE REQUEST TO INSTALL A LIBRARY DEPENDENT UPON THE NEW VERSION
354
DOWNLOAD THE LIBRARY WHOSE REQUEST FOR INSTALLATION WAS RECEIVED IN STEP 352
356
INSTALL THE LIBRARY DOWNLOADED IN STEP 354
358
USE THE LIBRARY INSTALLED IN STEP 356 WITH AN ASSURANCE OF A COMPATIBILITY BETWEEN THE INSTALLED LIBRARY AND DEPENDENT LIBRARIES AND WITH A MINIMIZED LIKELIHOOD OF DEPENDENCY CONFLICTS
360
END

*FIG. 3C*

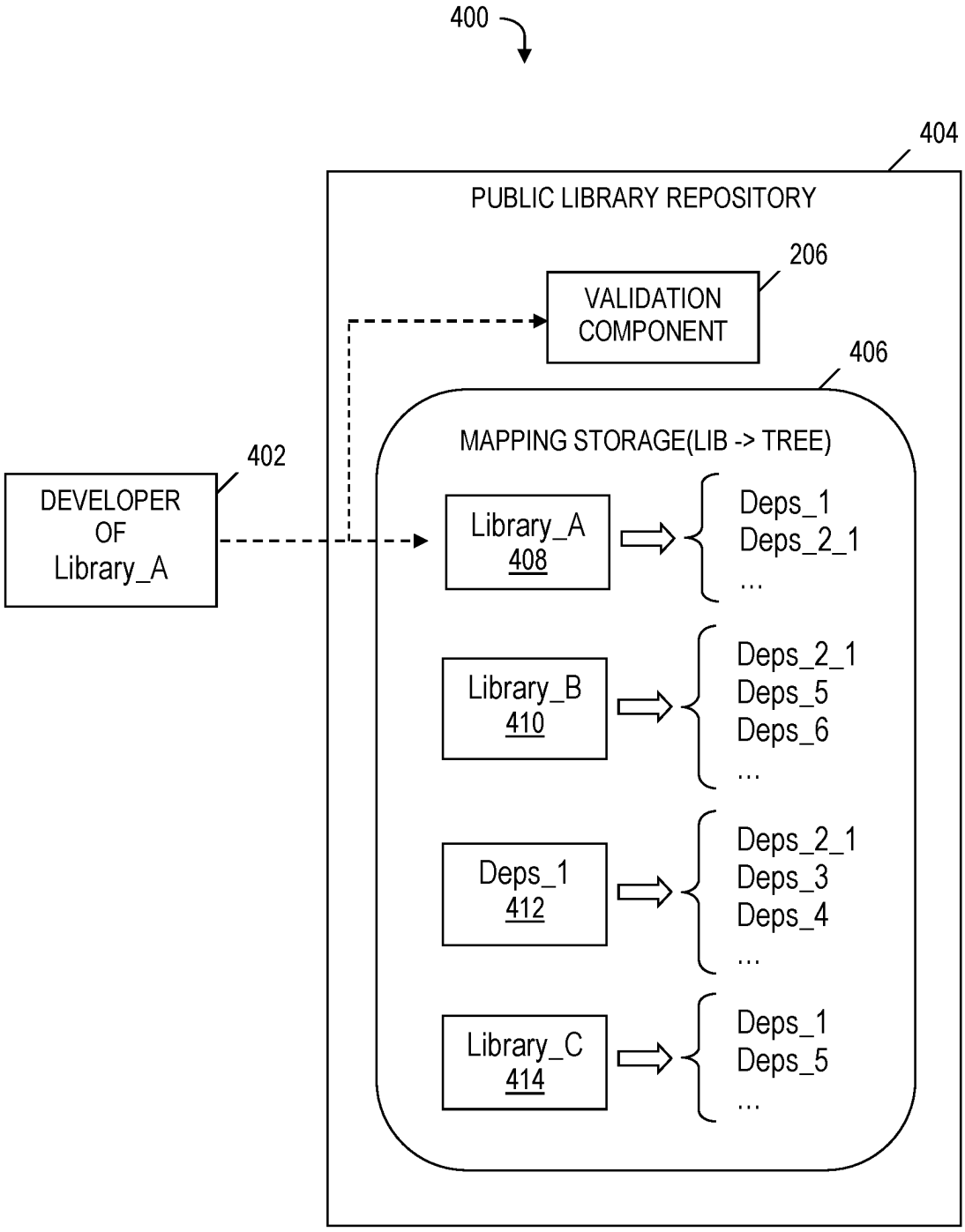

400
404
PUBLIC LIBRARY REPOSITORY
206
VALIDATION COMPONENT
406
MAPPING STORAGE(LIB -> TREE)
402
DEVELOPER OF Library_A
Library_A 408
Deps_1
Deps_2_1
...
Library_B 410
Deps_2_1
Deps_5
Deps_6
...
Deps_1 412
Deps_2_1
Deps_3
Deps_4
...
Library_C 414
Deps_1
Deps_5
...

*FIG. 4A*

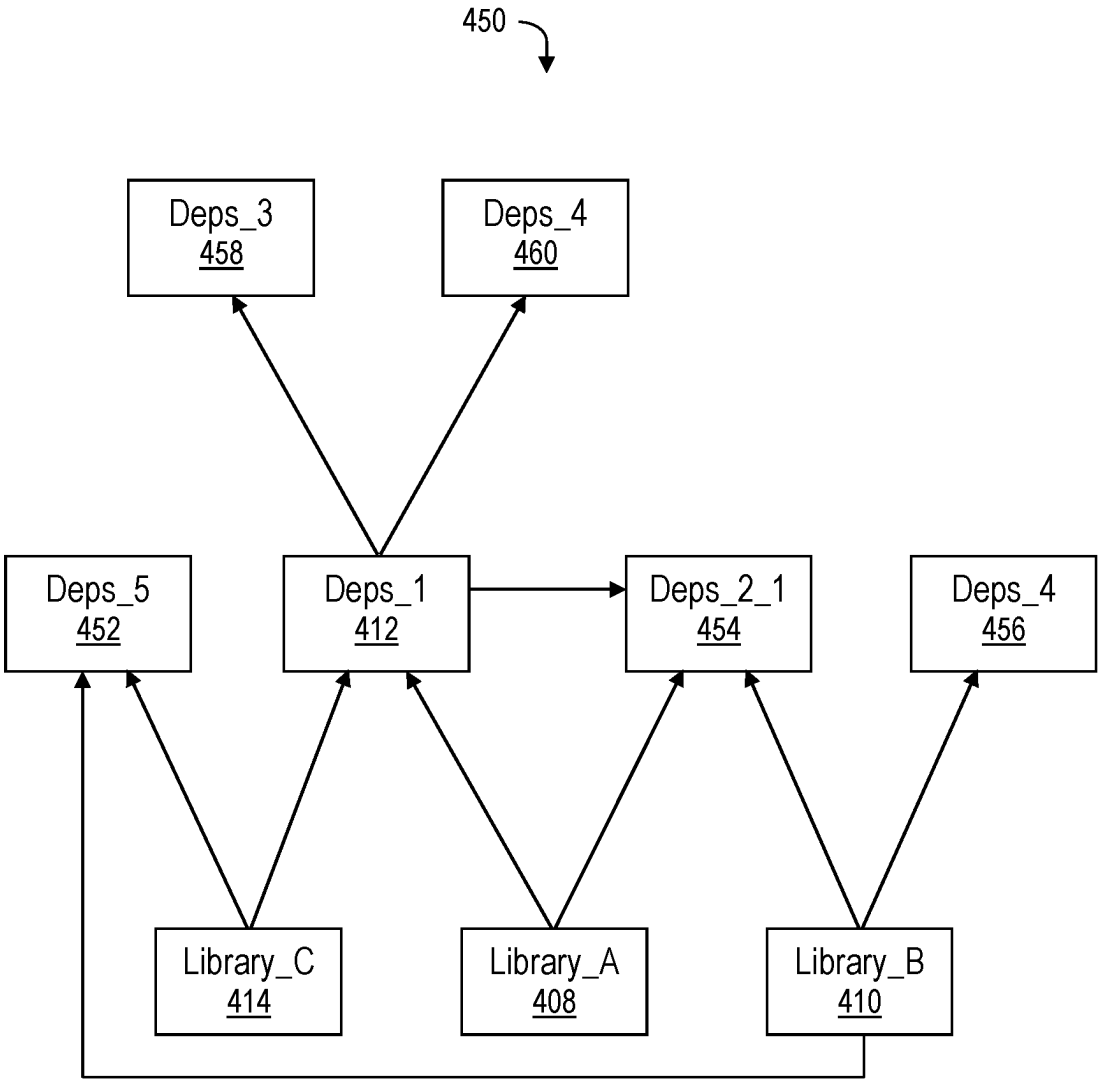

450
Deps_3
458
Deps_4
460
Deps_5
452
Deps_1
412
Deps_2_1
454
Deps_4
456
Library_C
414
Library_A
408
Library_B
410

*FIG. 4B*

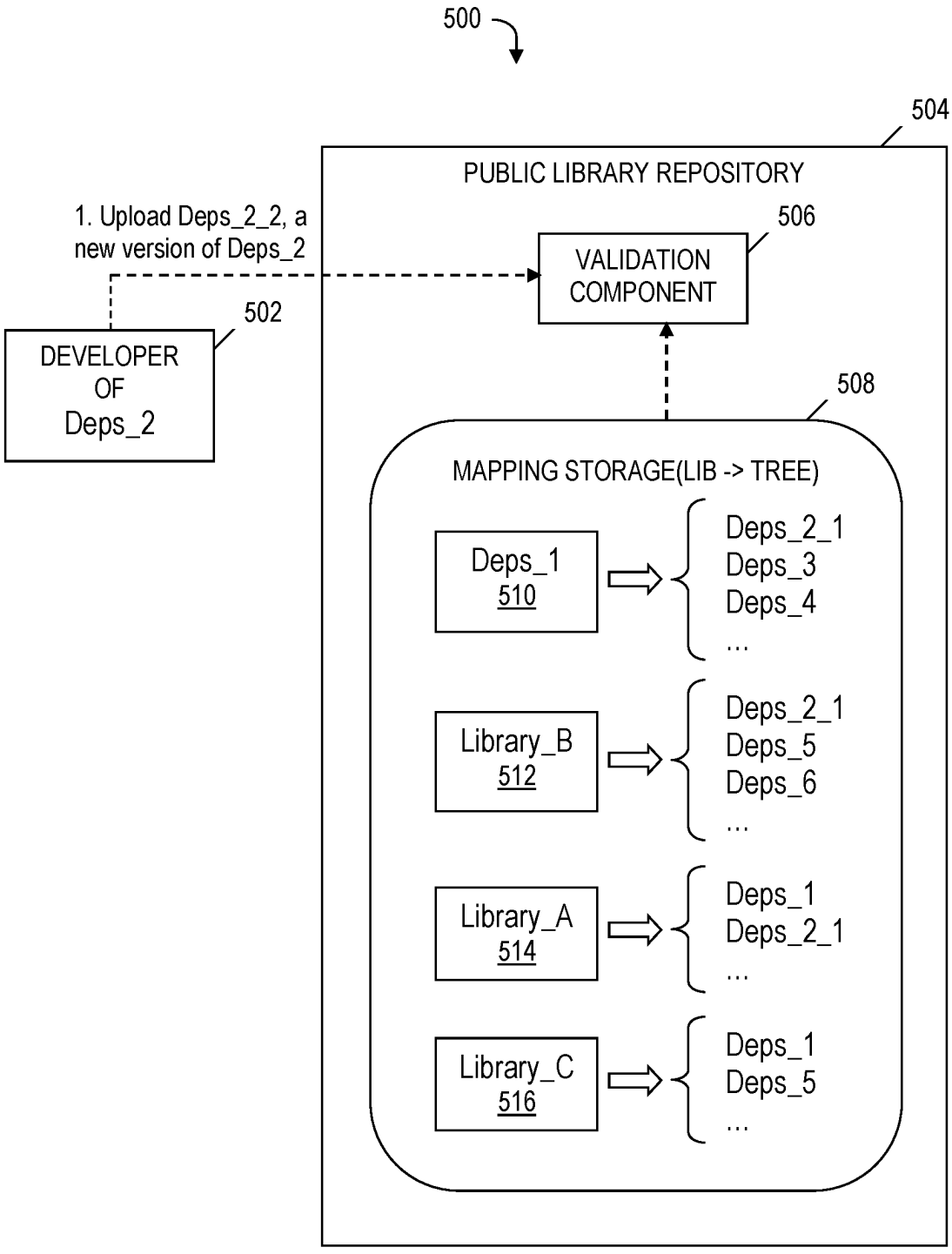

500
504
PUBLIC LIBRARY REPOSITORY
1. Upload Deps_2_2, a new version of Deps_2
506
VALIDATION COMPONENT
502
DEVELOPER OF Deps_2
508
MAPPING STORAGE(LIB -> TREE)
Deps_1
510
Deps_2_1
Deps_3
Deps_4
...
Library_B
512
Deps_2_1
Deps_5
Deps_6
...
Library_A
514
Deps_1
Deps_2_1
...
Library_C
516
Deps_1
Deps_5
...

*FIG. 5A*

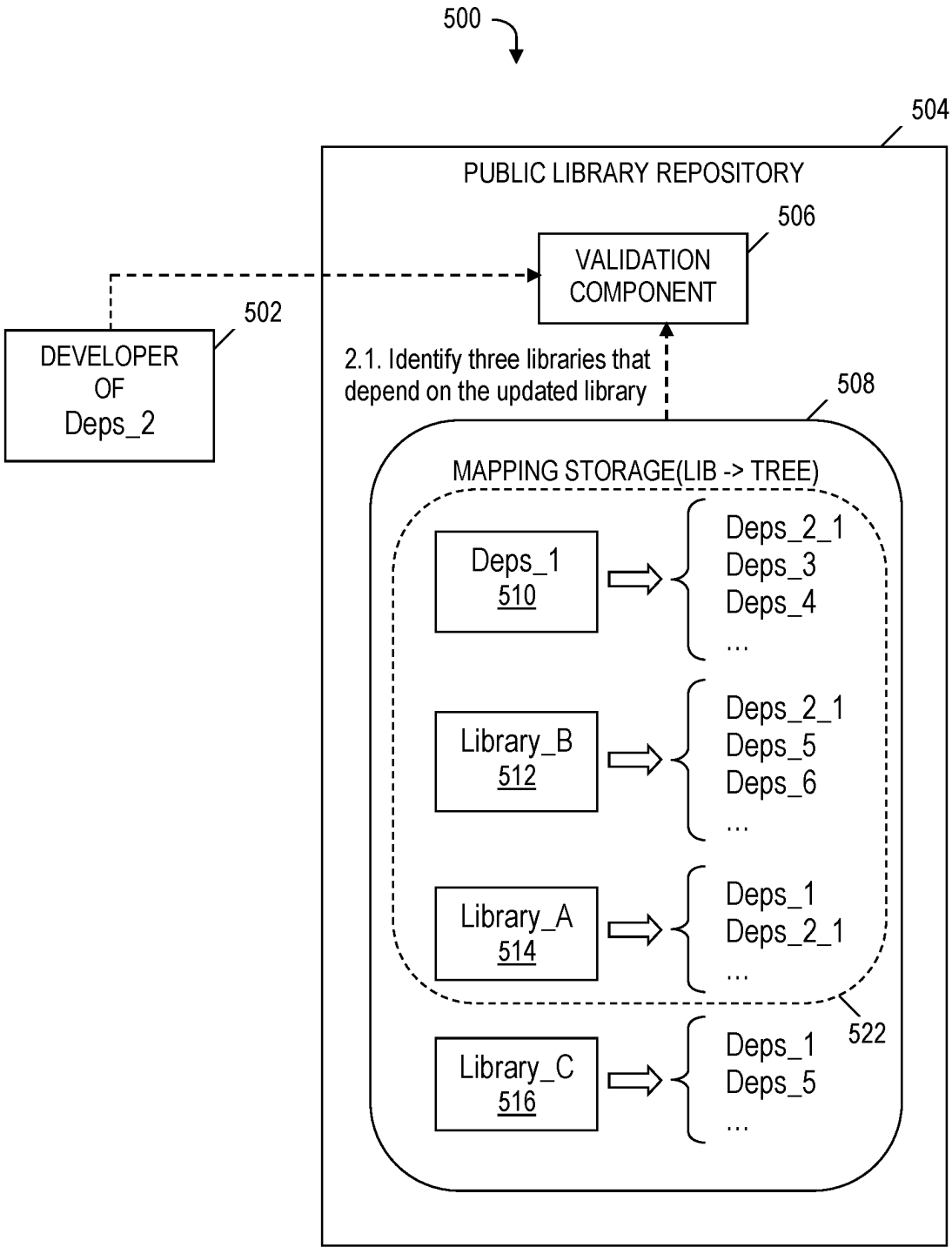

500
504
PUBLIC LIBRARY REPOSITORY
506
VALIDATION COMPONENT
502
DEVELOPER OF Deps_2
2.1. Identify three libraries that depend on the updated library
508
MAPPING STORAGE(LIB -> TREE)
Deps_1
510
Deps_2_1
Deps_3
Deps_4
...
Library_B
512
Deps_2_1
Deps_5
Deps_6
...
Library_A
514
Deps_1
Deps_2_1
...
522
Library_C
516
Deps_1
Deps_5
...

*FIG. 5B*

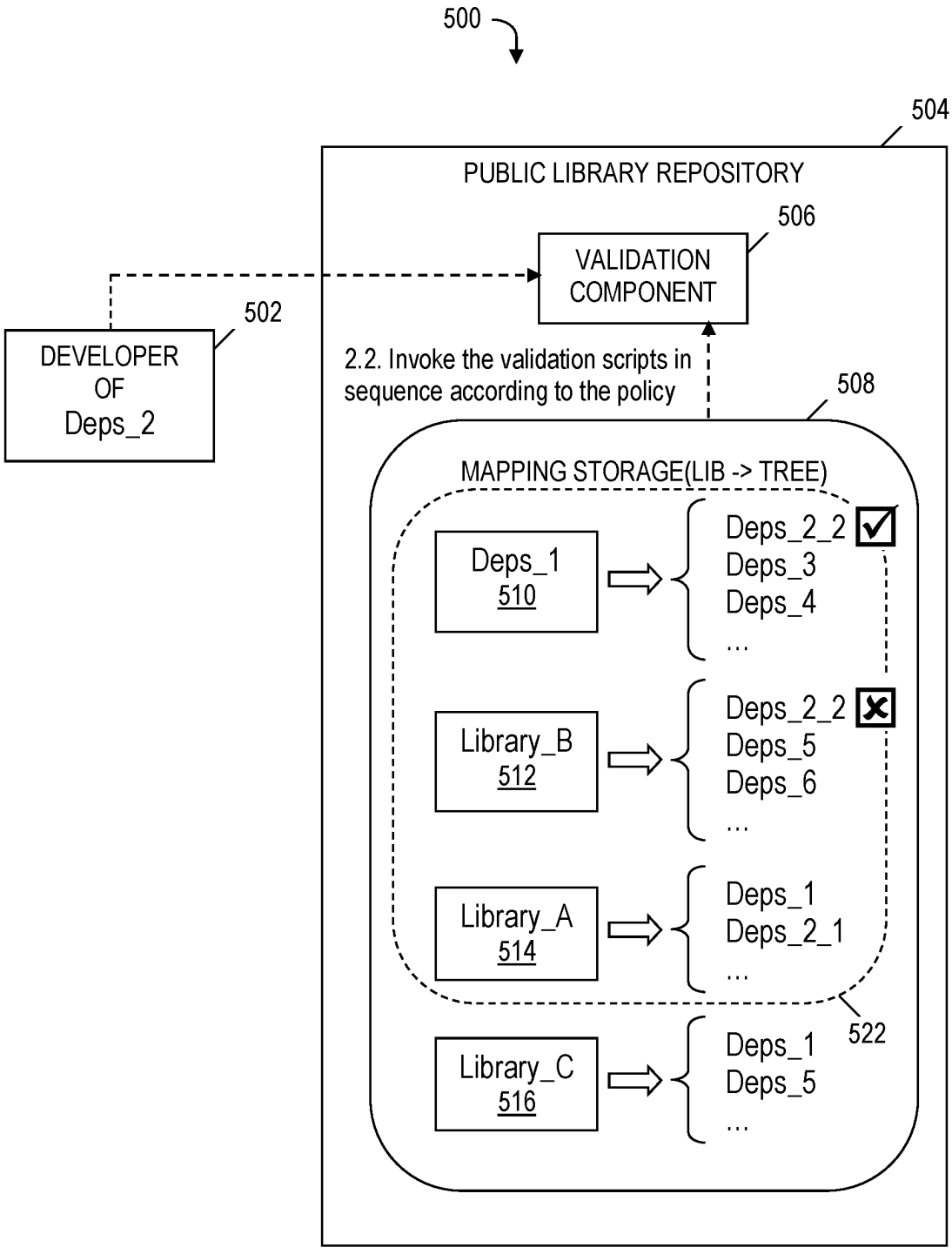

500
504
PUBLIC LIBRARY REPOSITORY
506
VALIDATION COMPONENT
502
DEVELOPER OF Deps_2
2.2. Invoke the validation scripts in sequence according to the policy
508
MAPPING STORAGE(LIB -> TREE)
Deps_1
510
Deps_2_2
Deps_3
Deps_4
...
Library_B
512
Deps_2_2
Deps_5
Deps_6
...
Library_A
514
Deps_1
Deps_2_1
...
522
Library_C
516
Deps_1
Deps_5
...

*FIG. 5C*

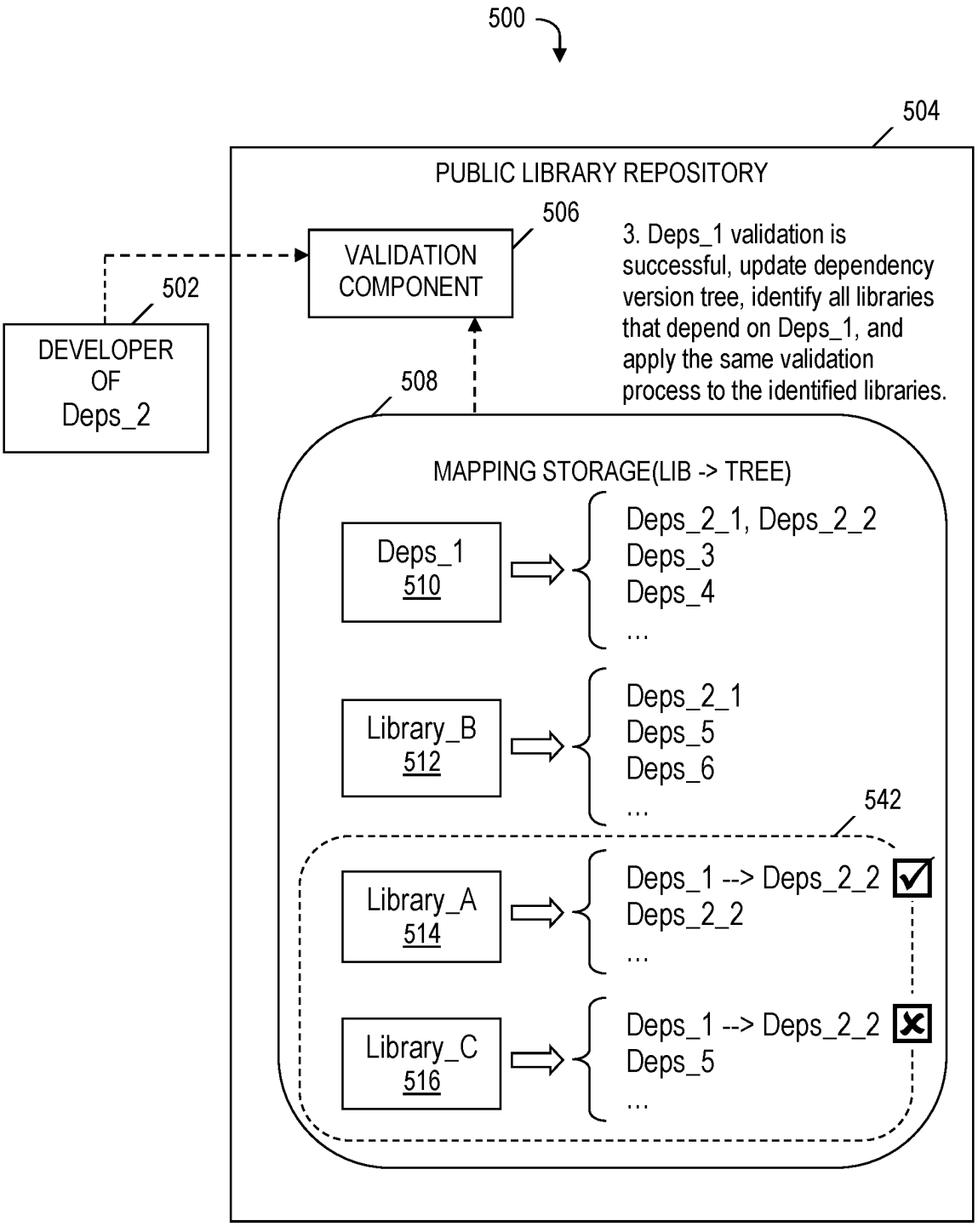

500
504
PUBLIC LIBRARY REPOSITORY
506
VALIDATION COMPONENT
3. Deps_1 validation is successful, update dependency version tree, identify all libraries that depend on Deps_1, and apply the same validation process to the identified libraries.
502
DEVELOPER OF Deps_2
508
MAPPING STORAGE(LIB -> TREE)
Deps_1
510
Deps_2_1, Deps_2_2
Deps_3
Deps_4
…
Library_B
512
Deps_2_1
Deps_5
Deps_6
…
542
Library_A
514
Deps_1 --> Deps_2_2
Deps_2_2
…
Library_C
516
Deps_1 --> Deps_2_2
Deps_5
…

*FIG. 5D*

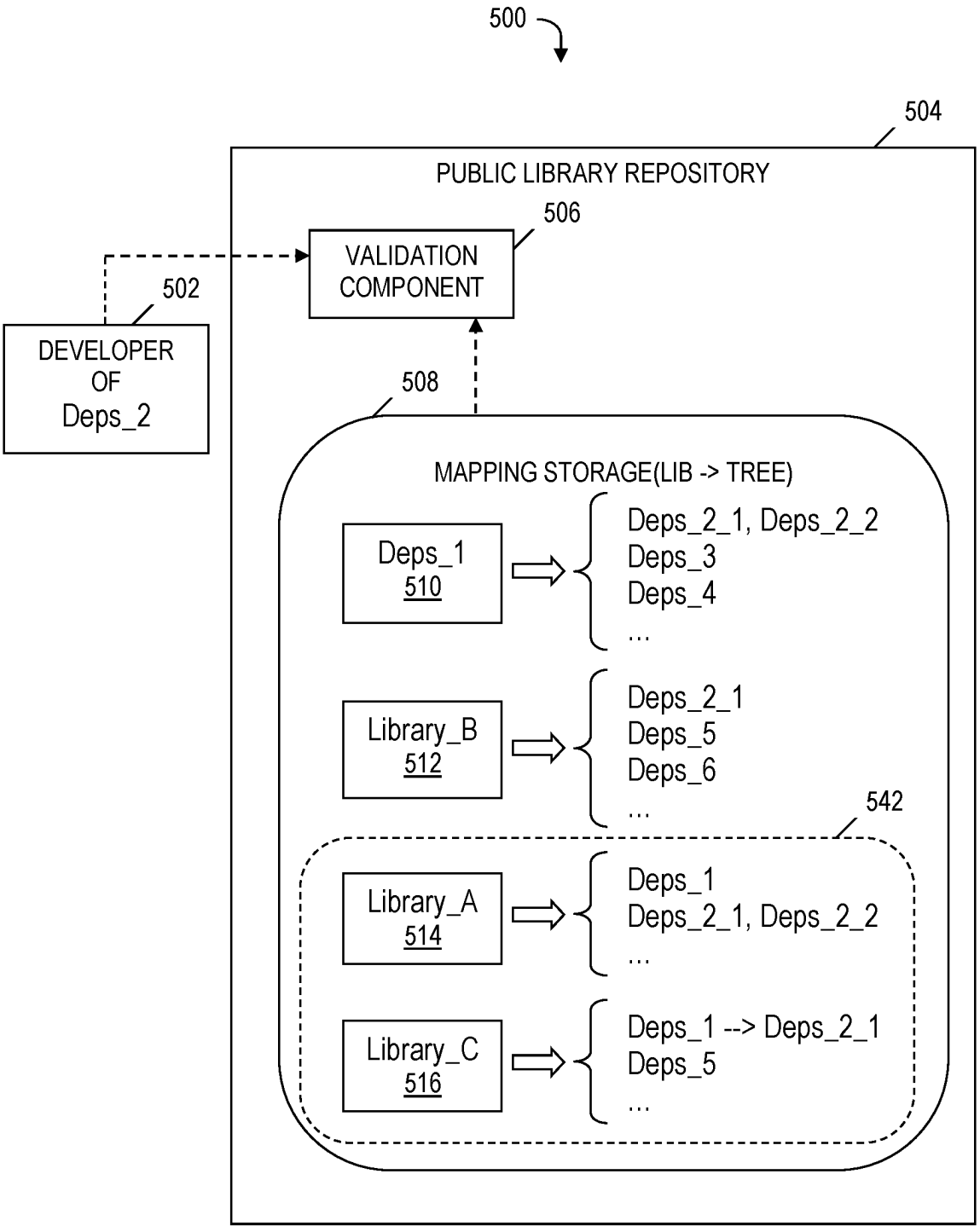

500
504
PUBLIC LIBRARY REPOSITORY
506
VALIDATION COMPONENT
502
DEVELOPER OF Deps_2
508
MAPPING STORAGE(LIB -> TREE)
Deps_1
510
Deps_2_1, Deps_2_2
Deps_3
Deps_4
...
Library_B
512
Deps_2_1
Deps_5
Deps_6
...
542
Library_A
514
Deps_1
Deps_2_1, Deps_2_2
...
Library_C
516
Deps_1 --> Deps_2_1
Deps_5
...

*FIG. 5E*

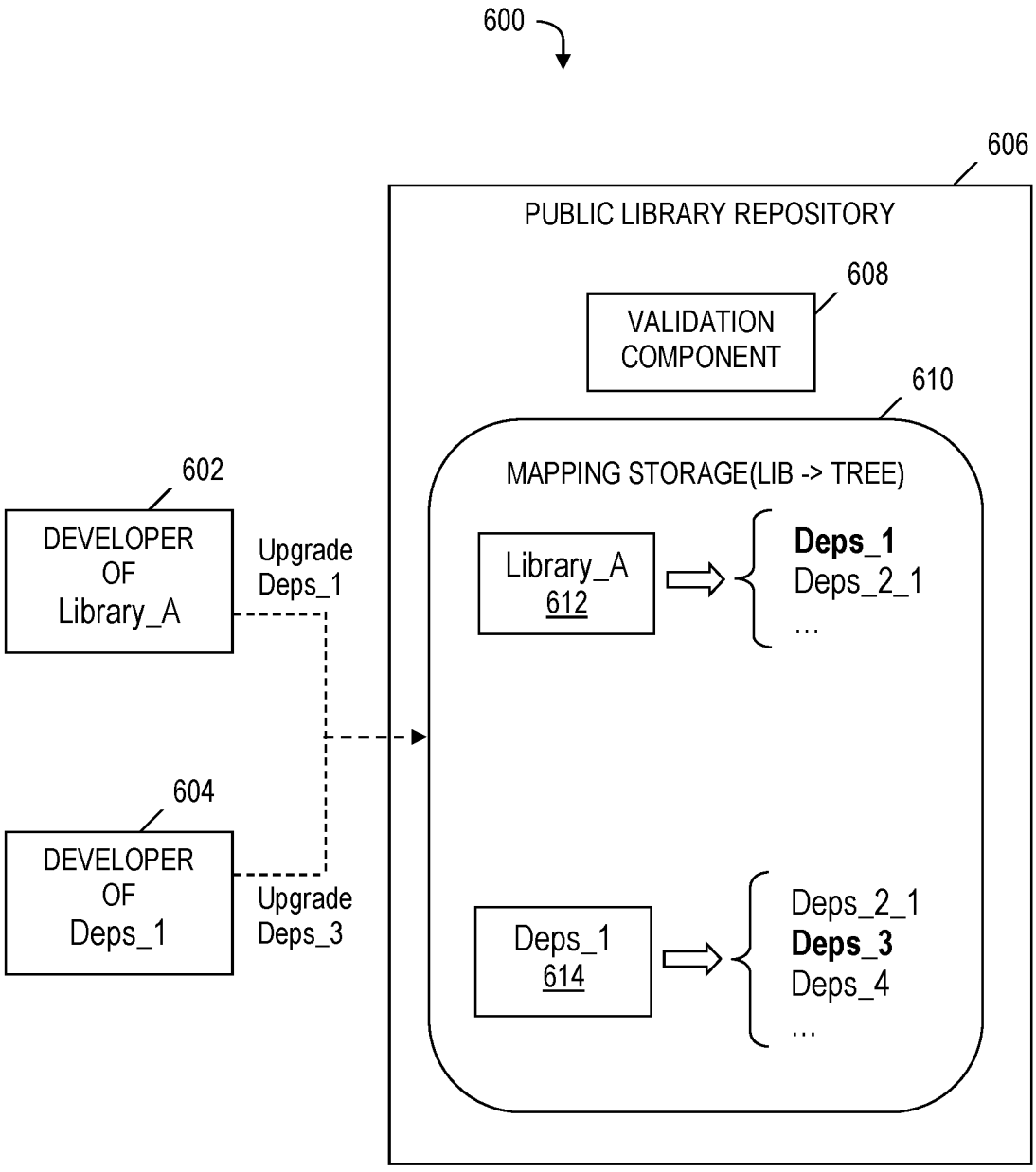

600
606
PUBLIC LIBRARY REPOSITORY
608
VALIDATION COMPONENT
610
MAPPING STORAGE(LIB -> TREE)
Library_A
612
Deps_1
Deps_2_1
...
602
DEVELOPER OF Library_A
Upgrade Deps_1
604
DEVELOPER OF Deps_1
Upgrade Deps_3
Deps_1
614
Deps_2_1
Deps_3
Deps_4
...

*FIG. 6*

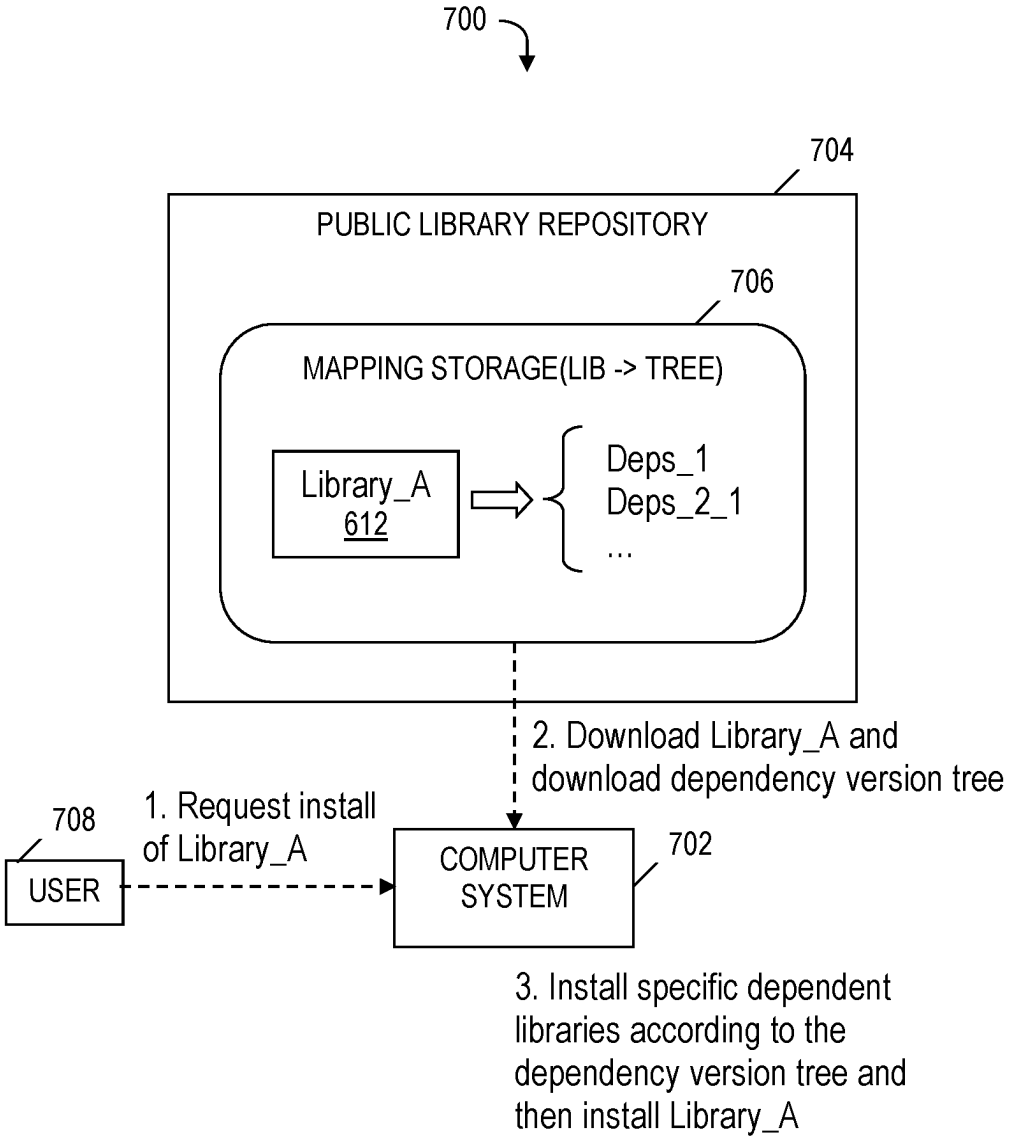

700
704
PUBLIC LIBRARY REPOSITORY
706
MAPPING STORAGE(LIB -> TREE)
Library_A
612
Deps_1
Deps_2_1
...
2. Download Library_A and
download dependency version tree
708
USER
1. Request install
of Library_A
COMPUTER
SYSTEM
702
3. Install specific dependent
libraries according to the
dependency version tree and
then install Library_A

*FIG. 7*

# LIBRARY DEVELOPMENT ENSURING LIBRARY COMPATIBILITY AND MINIMIZING DEPENDENCY CONFLICTS

## BACKGROUND

**[0001]** The present invention relates to software development, and more particularly to software library development.

## SUMMARY

**[0002]** In one embodiment, the present invention provides a computer-implemented method. The method includes receiving, by a processor set, requests to publish multiple libraries to a public library repository. The requests include specifications of respective sets of one or more dependencies of the multiple libraries and respective validation scripts. The method further includes building, by the processor set, a dependency version tree that includes the specifications of the respective sets of one or more dependencies. The method further includes receiving, by the processor set, a request to upload a new version of a library to the public library repository. The library is included in the multiple libraries. The method further includes validating, by a validation component and using the dependency version tree and the validation scripts, a compatibility between the library and one or more libraries dependent upon the library.

**[0003]** A computer system and a computer program product corresponding to the above-summarized computer-implemented method are also described herein.

## BRIEF DESCRIPTION OF THE DRAWINGS

**[0004]** FIG. **1** is a block diagram of a system for library development ensuring library compatibility and minimizing dependency conflicts, in accordance with embodiments of the present invention.

**[0005]** FIG. **2** is a block diagram of modules included in code included in the system of FIG. **1**, in accordance with embodiments of the present invention.

**[0006]** FIGS. **3**A-**3**B depict a flowchart of a process of library development ensuring library compatibility and minimizing dependency conflicts, where operations of the flowchart are performed by modules in FIG. **2**, in accordance with embodiments of the present invention.

**[0007]** FIG. **3**C is a flowchart of a process of installing a library in a local development environment, where the installation is performed by a module in FIG. **2**, in accordance with embodiments of the present invention.

**[0008]** FIGS. **4**A-**4**B depict an example of uploading a library and building a dependency version tree in the process of FIGS. **3**A-**3**B, in accordance with embodiments of the present invention.

**[0009]** FIGS. **5**A-**5**E depict an example of uploading a new version of a library and validating library compatibility by using a validation component and a dependency version tree in the process of FIGS. **3**A-**3**B, in accordance with embodiments of the present invention.

**[0010]** FIG. **6** is an example of queueing different updates of library dependencies based on respective positions in the dependency version tree, in accordance with embodiments of the present invention.

**[0011]** FIG. **7** is an example of installing a library and library dependencies in the process of FIG. **3**C, in accordance with embodiments of the present invention.

## DETAILED DESCRIPTION

### Overview

**[0012]** For a software development project, a developer often installs multiple libraries and manages the dependencies between libraries. The installation of the libraries can lead to dependency conflicts, where multiple libraries require different versions of the same library or where two libraries have dependencies on a third library that is not compatible with both of the two libraries. Dependency conflicts cause a range of issues, such as errors and crashes, making it difficult to develop and maintain projects. Resolving these conflicts requires significant time and effort, as a developer may need to manually adjust dependencies and versioning to ensure compatibility. Despite the use of conventional approaches to avoid dependency conflicts in actual development, the known management of dependencies in development is a complex task.

**[0013]** Under conventional approaches for managing dependencies, different projects may use inconsistent ways to declare dependencies, such as using setup.py or pyproject.toml, which lead to a lack of strong constraints on the referencing of the dependencies in a project's specific build environment. This lack of strong constraints results in unpredictable errors that are resolved only through manual debugging. Besides the inconsistency of dependency declaration file formats, for each dependency statement within the file, both a specific version and a version ranged specification are used, which leads to future conflicts when a new library version is published.

**[0014]** As one example of the aforementioned incompatibility issue arising under conventional approaches, a project is using two libraries, Library_A and Deps_1, both of which depend on library Deps_2. The latest version of Deps_2 is Deps_2_1. Under version Deps_2_1, both Library_A and Deps_1 run successfully and without any conflicts. Subsequently, a problem arises and the project stops working. After spending a significant amount of time and effort in researching the problem and manually debugging, it is discovered that Deps_2 had released a new version, Deps_2_2, and both Library_A and Deps_1 had automatically installed the new version, Deps_2_2. Deps_1, however, supports Deps_2_1, but not Deps_2_2, thereby causing conflicts and errors in the project.

**[0015]** Embodiments of the present invention address the aforementioned unique challenges by providing an approach that establishes a set of guidelines for library developers to follow so that dependency conflicts are prevented in the future. Embodiments of the present invention provide a standardized approach to library development that ensures compatibility with other libraries and minimizes the likelihood of dependency conflicts.

**[0016]** Embodiments of the present invention define a dependency version tree, which requires a library developer to submit a version range of all dependencies when publishing a new version of a library to a public library repository, such as Python® Package Index (PyPI®). Python and PyPI are registered trademarks of Python Software Foundation located in Beaverton, Oregon. Embodiments of the present invention provide novel features discussed herein that are language-neutral and are not limited to the Python® language. The requirement to submit the version range of all the dependencies when publishing a new library version provides a clear understanding of the dependencies

required by each library, and how the dependencies may conflict with other dependencies in a given project.

**[0017]** Embodiments of the present invention provide a validation component that requires a library developer to provide a custom validation script as part of the developer's library. When a developer updates the developer's library, the validation component automatically uses the maintained dependency version tree to find all libraries that depend on the updated library and invokes the validation script of the developer's library to verify library compatibility. If the validation is successful, the dependency version tree is updated and the validation recursively proceeds to verify the compatibility of the dependencies of one or more other libraries in the direction of the root node of the dependency version tree. The validation component ensures that libraries are properly validated and reduces the likelihood of errors and conflicts caused by inconsistent dependency versions.

**[0018]** The standardized approach to managing dependencies and ensuring compatibility disclosed herein reduces the occurrence of dependency conflicts in software development, thereby making it easier for developers to create high-quality software and maintain their software development projects over time.

Computing Environment

**[0019]** Various aspects of the present disclosure are described by narrative text, flowcharts, block diagrams of computer systems and/or block diagrams of the machine logic included in computer program product (CPP) embodiments. With respect to any flowcharts, depending upon the technology involved, the operations can be performed in a different order than what is shown in a given flowchart. For example, again depending upon the technology involved, two operations shown in successive flowchart blocks may be performed in reverse order, as a single integrated step, concurrently, or in a manner at least partially overlapping in time.

**[0020]** A computer program product embodiment ("CPP embodiment" or "CPP") is a term used in the present disclosure to describe any set of one, or more, computer-readable storage media (also called "mediums") collectively included in a set of one, or more, storage devices, and that collectively include machine readable code corresponding to instructions and/or data for performing computer operations specified in a given CPP claim. A "storage device" is any tangible device that can retain and store instructions for use by a computer processor. Without limitation, the computer-readable storage medium may be an electronic storage medium, a magnetic storage medium, an optical storage medium, an electromagnetic storage medium, a semiconductor storage medium, a mechanical storage medium, or any suitable combination of the foregoing. Some known types of storage devices that include these mediums include: diskette, hard disk, random access memory (RAM), read-only memory (ROM), erasable programmable read-only memory (EPROM or Flash memory), static random access memory (SRAM), compact disc read-only memory (CD-ROM), digital versatile disk (DVD), memory stick, floppy disk, mechanically encoded device (such as punch cards or pits/lands formed in a major surface of a disc) or any suitable combination of the foregoing. A computer-readable storage medium, as that term is used in the present disclosure, is not to be construed as storage in the form of transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide, light pulses passing through a fiber optic cable, electrical signals communicated through a wire, and/or other transmission media. As will be understood by those of skill in the art, data is typically moved at some occasional points in time during normal operations of a storage device, such as during access, de-fragmentation or garbage collection, but this does not render the storage device as transitory because the data is not transitory while it is stored.

**[0021]** FIG. **1** is a block diagram of a system for library development ensuring library compatibility and minimizing dependency conflicts, in accordance with embodiments of the present invention. Computing environment **100** contains an example of an environment for the execution of at least some of the computer code involved in performing the inventive methods, such as code **200** for library development ensuring library compatibility and minimizing dependency conflicts. The aforementioned computer code is also referred to herein as computer-readable code, computer-readable program code, and machine readable code. In addition to block **200**, computing environment **100** includes, for example, computer **101**, wide area network (WAN) **102**, end user device (EUD) **103**, remote server **104**, public cloud **105**, and private cloud **106**. In this embodiment, computer **101** includes processor set **110** (including processing circuitry **120** and cache **121**), communication fabric **111**, volatile memory **112**, persistent storage **113** (including operating system **122** and block **200**, as identified above), peripheral device set **114** (including user interface (UI) device set **123**, storage **124**, and Internet of Things (IoT) sensor set **125**), and network module **115**. Remote server **104** includes remote database **130**. Public cloud **105** includes gateway **140**, cloud orchestration module **141**, host physical machine set **142**, virtual machine set **143**, and container set **144**.

**[0022]** COMPUTER **101** may take the form of a desktop computer, laptop computer, tablet computer, smart phone, smart watch or other wearable computer, mainframe computer, quantum computer or any other form of computer or mobile device now known or to be developed in the future that is capable of running a program, accessing a network or querying a database, such as remote database **130**. As is well understood in the art of computer technology, and depending upon the technology, performance of a computer-implemented method may be distributed among multiple computers and/or between multiple locations. On the other hand, in this presentation of computing environment **100**, detailed discussion is focused on a single computer, specifically computer **101**, to keep the presentation as simple as possible. Computer **101** may be located in a cloud, even though it is not shown in a cloud in FIG. **1**. On the other hand, computer **101** is not required to be in a cloud except to any extent as may be affirmatively indicated.

**[0023]** PROCESSOR SET **110** includes one, or more, computer processors of any type now known or to be developed in the future. Processing circuitry **120** may be distributed over multiple packages, for example, multiple, coordinated integrated circuit chips. Processing circuitry **120** may implement multiple processor threads and/or multiple processor cores. Cache **121** is memory that is located in the processor chip package(s) and is typically used for data or code that should be available for rapid access by the threads or cores running on processor set **110**. Cache memo-

ries are typically organized into multiple levels depending upon relative proximity to the processing circuitry. Alternatively, some, or all, of the cache for the processor set may be located "off chip." In some computing environments, processor set **110** may be designed for working with qubits and performing quantum computing.

**[0024]** Computer-readable program instructions are typically loaded onto computer **101** to cause a series of operational steps to be performed by processor set **110** of computer **101** and thereby effect a computer-implemented method, such that the instructions thus executed will instantiate the methods specified in flowcharts and/or narrative descriptions of computer-implemented methods included in this document (collectively referred to as "the inventive methods"). These computer-readable program instructions are stored in various types of computer-readable storage media, such as cache **121** and the other storage media discussed below. The program instructions, and associated data, are accessed by processor set **110** to control and direct performance of the inventive methods. In computing environment **100**, at least some of the instructions for performing the inventive methods may be stored in block **200** in persistent storage **113**.

**[0025]** COMMUNICATION FABRIC **111** is the signal conduction path that allows the various components of computer **101** to communicate with each other. Typically, this fabric is made of switches and electrically conductive paths, such as the switches and electrically conductive paths that make up busses, bridges, physical input/output ports and the like. Other types of signal communication paths may be used, such as fiber optic communication paths and/or wireless communication paths.

**[0026]** VOLATILE MEMORY **112** is any type of volatile memory now known or to be developed in the future. Examples include dynamic type random access memory (RAM) or static type RAM. Typically, volatile memory **112** is characterized by random access, but this is not required unless affirmatively indicated. In computer **101**, the volatile memory **112** is located in a single package and is internal to computer **101**, but, alternatively or additionally, the volatile memory may be distributed over multiple packages and/or located externally with respect to computer **101**.

**[0027]** PERSISTENT STORAGE **113** is any form of non-volatile storage for computers that is now known or to be developed in the future. The non-volatility of this storage means that the stored data is maintained regardless of whether power is being supplied to computer **101** and/or directly to persistent storage **113**. Persistent storage **113** may be a read only memory (ROM), but typically at least a portion of the persistent storage allows writing of data, deletion of data and re-writing of data. Some familiar forms of persistent storage include magnetic disks and solid state storage devices. Operating system **122** may take several forms, such as various known proprietary operating systems or open source Portable Operating System Interface-type operating systems that employ a kernel. The code included in block **200** typically includes at least some of the computer code involved in performing the inventive methods.

**[0028]** PERIPHERAL DEVICE SET **114** includes the set of peripheral devices of computer **101**. Data communication connections between the peripheral devices and the other components of computer **101** may be implemented in various ways, such as Bluetooth connections, Near-Field Communication (NFC) connections, connections made by cables (such as universal serial bus (USB) type cables), insertion-type connections (for example, secure digital (SD) card), connections made through local area communication networks and even connections made through wide area networks such as the internet. In various embodiments, UI device set **123** may include components such as a display screen, speaker, microphone, wearable devices (such as goggles and smart watches), keyboard, mouse, printer, touchpad, game controllers, and haptic devices. Storage **124** is external storage, such as an external hard drive, or insertable storage, such as an SD card. Storage **124** may be persistent and/or volatile. In some embodiments, storage **124** may take the form of a quantum computing storage device for storing data in the form of qubits. In embodiments where computer **101** is required to have a large amount of storage (for example, where computer **101** locally stores and manages a large database) then this storage may be provided by peripheral storage devices designed for storing very large amounts of data, such as a storage area network (SAN) that is shared by multiple, geographically distributed computers. IoT sensor set **125** is made up of sensors that can be used in Internet of Things applications. For example, one sensor may be a thermometer and another sensor may be a motion detector.

**[0029]** NETWORK MODULE **115** is the collection of computer software, hardware, and firmware that allows computer **101** to communicate with other computers through WAN **102**. Network module **115** may include hardware, such as modems or Wi-Fi signal transceivers, software for packetizing and/or de-packetizing data for communication network transmission, and/or web browser software for communicating data over the internet. In some embodiments, network control functions and network forwarding functions of network module **115** are performed on the same physical hardware device. In other embodiments (for example, embodiments that utilize software-defined networking (SDN)), the control functions and the forwarding functions of network module **115** are performed on physically separate devices, such that the control functions manage several different network hardware devices. Computer-readable program instructions for performing the inventive methods can typically be downloaded to computer **101** from an external computer or external storage device through a network adapter card or network interface included in network module **115**.

**[0030]** WAN **102** is any wide area network (for example, the internet) capable of communicating computer data over non-local distances by any technology for communicating computer data, now known or to be developed in the future. In some embodiments, the WAN **102** may be replaced and/or supplemented by local area networks (LANs) designed to communicate data between devices located in a local area, such as a Wi-Fi network. The WAN and/or LANs typically include computer hardware such as copper transmission cables, optical transmission fibers, wireless transmission, routers, firewalls, switches, gateway computers and edge servers.

**[0031]** END USER DEVICE (EUD) **103** is any computer system that is used and controlled by an end user (for example, a customer of an enterprise that operates computer **101**), and may take any of the forms discussed above in connection with computer **101**. EUD **103** typically receives helpful and useful data from the operations of computer **101**. For example, in a hypothetical case where computer **101** is

designed to provide a recommendation to an end user, this recommendation would typically be communicated from network module **115** of computer **101** through WAN **102** to EUD **103**. In this way, EUD **103** can display, or otherwise present, the recommendation to an end user. In some embodiments, EUD **103** may be a client device, such as thin client, heavy client, mainframe computer, desktop computer and so on.

**[0032]** REMOTE SERVER **104** is any computer system that serves at least some data and/or functionality to computer **101**. Remote server **104** may be controlled and used by the same entity that operates computer **101**. Remote server **104** represents the machine(s) that collect and store helpful and useful data for use by other computers, such as computer **101**. For example, in a hypothetical case where computer **101** is designed and programmed to provide a recommendation based on historical data, then this historical data may be provided to computer **101** from remote database **130** of remote server **104**.

**[0033]** PUBLIC CLOUD **105** is any computer system available for use by multiple entities that provides on-demand availability of computer system resources and/or other computer capabilities, especially data storage (cloud storage) and computing power, without direct active management by the user. Cloud computing typically leverages sharing of resources to achieve coherence and economies of scale. The direct and active management of the computing resources of public cloud **105** is performed by the computer hardware and/or software of cloud orchestration module **141**. The computing resources provided by public cloud **105** are typically implemented by virtual computing environments that run on various computers making up the computers of host physical machine set **142**, which is the universe of physical computers in and/or available to public cloud **105**. The virtual computing environments (VCEs) typically take the form of virtual machines from virtual machine set **143** and/or containers from container set **144**. It is understood that these VCEs may be stored as images and may be transferred among and between the various physical machine hosts, either as images or after instantiation of the VCE. Cloud orchestration module **141** manages the transfer and storage of images, deploys new instantiations of VCEs and manages active instantiations of VCE deployments. Gateway **140** is the collection of computer software, hardware, and firmware that allows public cloud **105** to communicate through WAN **102**.

**[0034]** Some further explanation of virtualized computing environments (VCEs) will now be provided. VCEs can be stored as "images." A new active instance of the VCE can be instantiated from the image. Two familiar types of VCEs are virtual machines and containers. A container is a VCE that uses operating-system-level virtualization. This refers to an operating system feature in which the kernel allows the existence of multiple isolated user-space instances, called containers. These isolated user-space instances typically behave as real computers from the point of view of programs running in them. A computer program running on an ordinary operating system can utilize all resources of that computer, such as connected devices, files and folders, network shares, CPU power, and quantifiable hardware capabilities. However, programs running inside a container can only use the contents of the container and devices assigned to the container, a feature which is known as containerization.

**[0035]** PRIVATE CLOUD **106** is similar to public cloud **105**, except that the computing resources are only available for use by a single enterprise. While private cloud **106** is depicted as being in communication with WAN **102**, in other embodiments a private cloud may be disconnected from the internet entirely and only accessible through a local/private network. A hybrid cloud is a composition of multiple clouds of different types (for example, private, community or public cloud types), often respectively implemented by different vendors. Each of the multiple clouds remains a separate and discrete entity, but the larger hybrid cloud architecture is bound together by standardized or proprietary technology that enables orchestration, management, and/or data/application portability between the multiple constituent clouds. In this embodiment, public cloud **105** and private cloud **106** are both part of a larger hybrid cloud.

**[0036]** CLOUD COMPUTING SERVICES AND/OR MICROSERVICES (not separately shown in FIG. **1**): private and public clouds **106** are programmed and configured to deliver cloud computing services and/or microservices (unless otherwise indicated, the word "microservices" shall be interpreted as inclusive of larger "services" regardless of size). Cloud services are infrastructure, platforms, or software that are typically hosted by third-party providers and made available to users through the internet. Cloud services facilitate the flow of user data from front-end clients (for example, user-side servers, tablets, desktops, laptops), through the internet, to the provider's systems, and back. In some embodiments, cloud services may be configured and orchestrated according to an "as a service" technology paradigm where something is being presented to an internal or external customer in the form of a cloud computing service. As-a-Service offerings typically provide endpoints with which various customers interface. These endpoints are typically based on a set of APIs. One category of as-a-service offering is Platform as a Service (PaaS), where a service provider provisions, instantiates, runs, and manages a modular bundle of code that customers can use to instantiate a computing platform and one or more applications, without the complexity of building and maintaining the infrastructure typically associated with these things. Another category is Software as a Service (SaaS) where software is centrally hosted and allocated on a subscription basis. SaaS is also known as on-demand software, web-based software, or web-hosted software. Four technological sub-fields involved in cloud services are: deployment, integration, on demand, and virtual private networks.

System and Process for Library Development Ensuring Library Compatibility and Minimizing Dependency Conflicts

**[0037]** FIG. **2** is a block diagram of modules included in code **200** included in system **100** of FIG. **1**, in accordance with embodiments of the present invention. Code **200** includes a dependency version tree build module **202**, a new library version upload module **204**, a validation component **206**, a dependency version tree update module **208**, and a library installation module **210**.

**[0038]** Dependency version tree build module **202** is configured to build a dependency version tree that includes specifications of multiple libraries and sets of one or more dependencies of the multiple libraries. Dependency version tree build module **202** is further configured to receive requests to publish the aforementioned multiple libraries to

a public library repository (e.g., PyPI®). A received request to publish a library includes a submission by a library developer of a version range of all of the one or more dependencies of the library. As used herein, a dependency of a given library is a library that depends on the given library. The received request to publish a library also includes a customized validation script as part of the library. The customized validation script verifies a compatibility between the library being published and the library's dependencies.

**[0039]** New library version upload module **204** is configured to receive a request to upload a new version of a library to the public library repository. New library version upload module **204** is further configured to upload the new version of the library to the public library repository.

**[0040]** Validation component **206** is configured to identify all of the one or more libraries dependent upon an uploaded library or an uploaded new version of a library and to verify whether or not there is compatibility between the uploaded library or new version of the library and each of the identified one or more libraries dependent upon the uploaded library or new version of the library. In one embodiment, validation component **206** is further configured to detect the upload of a new version of the library performed by the new library version upload module **204** and to be triggered in response to detecting the aforementioned upload of the new version.

**[0041]** Dependency version tree update module **208** is configured to update the dependency version tree in response to the verification by the validation component **206** being successful. The update of the dependency version tree includes adding the new version of the library in an extended range of versions to which a given library is mapped, where the given library is selected from the identified one or more libraries dependent upon the uploaded new version of the library, and where the selection of the given library is performed by validation component **206**.

**[0042]** Library installation module **210** is configured to (i) receive a request to install a library dependent upon another library (e.g., the new version of the library uploaded by the new library version upload module **204**); (ii) download the requested library from the public library repository; (iii) install the library whose installation was requested; and (iv) use the installed library with an assurance of compatibility between the installed library and one or more dependent libraries and with a minimized likelihood of dependency conflicts.

**[0043]** The functionality of the modules included in code **200** is described in more detail in the discussions presented below relative to FIGS. **3**A-**3**B, FIG. **3**C, FIGS. **4**A-**4**B, FIGS. **5**A-**5**E, FIG. **6**, and FIG. **7**.

**[0044]** FIGS. **3**A-**3**B depict a flowchart of a process of library development ensuring library compatibility and minimizing dependency conflicts, where operations of the flowchart are performed by modules in FIG. **2**, in accordance with embodiments of the present invention. The process of FIGS. **3**A-**3**B begins at a start node **300**. In step **302**, dependency version tree build module **202** receives requests to publish multiple libraries to a public library repository. The requests to publish include (i) specifications of respective sets of one or more dependencies of the multiple libraries and (ii) respective validation scripts. In one embodiment, a request to publish a library includes the library being published, along with (i) a declaration of all of the one or more dependencies of the library in a tree format, which indicates a version or a specified range of versions for each dependency and (ii) a custom script (also referred to herein as a validation script or a customized validation script) that validates the compatibility between the library being published and the library's one or more dependencies.

**[0045]** The submission of the range of versions when publishing a new version of a given library provides a clear understanding of the dependencies required by the given library and how a dependency may conflict with other dependencies in a given software development project.

**[0046]** In step **304**, dependency version tree build module **202** builds a dependency version tree that includes the specifications of the respective sets of the one or more dependencies of the multiple libraries.

**[0047]** In step **306**, new library version upload module **204** receives a request to upload a new version of a library to the public library repository. The library whose new version's upload is being requested is a library included in the aforementioned multiple libraries which were published in the public library repository in response to the requests received in step **302**.

**[0048]** In step **308**, new library version upload module **204** uploads the new version of the library to the public library repository.

**[0049]** In step **310**, validation component **206** detects the upload in step **308** and is triggered.

**[0050]** In step **312**, in response to detecting the upload in step **310**, validation component **206** identifies one or more libraries dependent upon the new version of the library uploaded in step **308**. The identification of the one or more libraries in step **312** utilizes the dependency version tree.

**[0051]** In step **314**, validation component **206** selects a first library included in the one or more libraries identified in step **312**. Alternatively, if step **314** is being repeated in a loop described below, then validation component **206** selects a next library included in the one or more libraries identified in step **312**. Hereinafter, in the discussion of FIGS. **3**A-**3**B, the first library or next library selected in the most recent performance of step **314** is referred to as the "selected library."

**[0052]** In step **316**, validation component **206** validates the selected library by verifying whether or not there is compatibility between the uploaded new version of the library and the selected library, where the verifying includes invoking a validation script associated with the selected library. The invoked validation script is included in the validation scripts that are included in the requests to publish discussed above relative to step **302**. In one embodiment, the validation in step **316** is performed in response to the detection of the upload of the new version of the library in step **310**.

**[0053]** Following step **314**, the process of FIGS. **3**A-**3**B continues with step **318** in FIG. **3**B.

**[0054]** In step **318**, validation component **206** determines whether the validation of the selected library is successful (i.e., determines whether the compatibility between the uploaded new version of the library and the selected library is verified). If validation component **206** determines in step **318** that the validation of the selected library is not successful, then the No branch of step **318** is followed, and the process of FIGS. **3**A-**3**B ends at an end node **319**. Returning to step **318**, if validation component **206** determines that the validation of the selected library is successful, then the Yes branch of step **318** is followed and step **320** is performed.

**[0055]** In step **320**, dependency version tree update module **208** updates the dependency version tree by adding the new version of the library in an extended range of versions to which the selected library is mapped.

**[0056]** In step **322**, validation component **206** recursively validates the dependencies of other libraries starting with a validation of all of the one or more libraries that depend on the selected library, which includes identifying the one or more libraries that depend on the selected library and then repeating the validation, the determination of a successful validation, and the updating of the dependency version tree in steps **316**, **318**, and **320**. The recursively validating in step **322** includes traversing a path in the dependency version tree by moving from a node of the dependency version tree associated with the selected library towards a root node of the dependency version tree. The aforementioned validation and recursive validation ensure that libraries are properly validated and reduces the likelihood of errors and conflicts caused by inconsistent dependency versions.

**[0057]** In step **324**, validation component **206** determines whether there is another library to select from the one or more libraries identified in step **312**. If validation component **206** determines in step **324** that there is another library (i.e., a next library) to select from the one or more libraries identified in step **312**, then the Yes branch of step **324** is followed, and the process of FIGS. **3**A-**3**B loops back to step **314** in FIG. **3**A, which selects the next library.

**[0058]** Returning to step **324**, if validation component **206** determines that there is not another library (i.e., there is not a next library) to select from the one or more libraries identified in step **312**, then the No branch of step **324** is followed and the process of FIGS. **3**A-**3**B ends at an end node **326**.

**[0059]** FIG. **3**C is a flowchart of a process of installing a library in a local development environment, where the installation is performed by a module in FIG. **2**, in accordance with embodiments of the present invention. The process of FIG. **3**C begins at a start node **350**. In step **352**, library installation module **210** receives a request from a user to install a library that is dependent upon the new version of the library that was uploaded in step **308** in FIG. **3**A. The request to install the library is received via a computer system utilized by the user.

**[0060]** In step **354**, library installation module **210** downloads the library whose request for installation was received in step **352**. The aforementioned library is downloaded in step **354** from the public library repository. Also in step **354**, library installation module **210** downloads the dependency version tree from the public library repository.

**[0061]** In step **356** and based on the downloaded dependency version tree, library installation module **210** installs one or more dependent libraries that are dependent upon the library downloaded in step **354**, and subsequently installs the library downloaded in step **354**. Step **356** includes installing the one or more dependent libraries and the library downloaded in step **354** onto the computer system utilized by the user.

**[0062]** In step **358**, the computer system utilized by the user uses the library installed in step **356** in software development action(s), where the use of the library is performed with an assurance of a compatibility of the installed library with any of the one or more dependent libraries and with a minimized likelihood of a conflict between the installed library and any one of the one or more dependent libraries.

**[0063]** Following step **358**, the process of FIG. **3**C ends at an end node **360**.

**[0064]** FIGS. **4**A-**4**B depict an example of uploading a library and building a dependency version tree in the process of FIGS. **3**A-**3**B, in accordance with embodiments of the present invention. Example **400** in FIG. **4**A includes a developer **402** of Library_A and a public library repository **404**, which includes a mapping storage **406** and validation component **206**. Developer **402** of Library_A submits a new library (i.e., Library_A) and a validation script, where the new library is added to public library repository **404**, and the validation script is stored in validation component **206**. The submitted validation script is a customized script that validates Library_A by verifying a compatibility between Library_A and another library on which Library_A depends. Mapping storage **406** includes mappings of (i) Library_A to dependencies that include Deps_1 and Deps_2_1; (ii) Library_B to dependencies that include Deps_2_1, Deps_5, and Deps_6; (iii) Deps_1 to dependencies that include Deps_2_1, Deps_3, and Deps_4; and (iv) Library_C to dependencies that include Deps_1 and Deps_5.

**[0065]** Dependency version tree build module **202** builds a dependency version tree **450** in FIG. **4**B from the information in mapping storage **406** in FIG. **4**A. Dependency version tree **450** is maintained by public library repository **404** rather than in the project of developer **402** of Library_A (or any other developer). Dependency version tree **450** specifies, for example, that Library_A is dependent on Deps_1 and Deps_2_1, and that Deps_1 is dependent on Deps_3, Deps_4, and Deps_2_1.

**[0066]** FIGS. **5**A-**5**E depict an example **500** of uploading a new version of a library and validating library compatibility by using a validation component and a dependency version tree in the process of FIGS. **3**A-**3**B, in accordance with embodiments of the present invention. Example **500** includes a developer **502** of Deps_2 and a public library repository **504**, which includes a validation component **506** and a mapping storage **508**. Public library repository **504** is an example of public library repository **404**, validation component is an example of validation component **206**, and mapping storage **508** is an example of mapping storage **406**.

**[0067]** In FIG. **5**A, example **500** includes a step 1, which includes developer **502** of Deps_2 uploading a new version of the library Deps_2 to public library repository **504**, where the old version of Deps_2 is Deps_2_1 and the new version of Deps_2 is Deps_2_2. The uploading of the new version, Deps_2_2, is an example of step **308** in FIG. **3**A. Validation component **506** detects the upload of Deps_2_2 and is triggered.

**[0068]** In FIG. **5**B, example **500** includes a step 2.1, which includes validation component **506** identifying three libraries **522** (i.e., Deps_1, Library_B, and Library_A) that depend on the library being updated (i.e., Deps_2). Validation component **506** identifies Deps_1, Library_B, and Library_A as all the libraries that depend on Deps_2_1, which is the old version of Deps_2. The identification of the three libraries is an example of step **312** in FIG. **3**A.

**[0069]** In FIG. **5**C, example **500** includes a step 2.2, which includes validation component **506** invoking the validation scripts of the three libraries **522**, which were identified in step 2.1 in FIG. **5**B to verify compatibility between Deps_

2_2 (i.e., the uploaded new version of library Deps_2) and each of the identified three libraries **522** (i.e., Deps_1, Library_B, and Library_A). Step 2.2 is an example of step **316** in FIG. **3**A. The validation scripts are invoked in a sequence according to a specified policy. The invocation of the validation scripts is an example of step **316** in FIG. **3**A.

**[0070]** The checkmark in FIG. **5**C indicates that running the validation script associated with Deps_1 verifies that Deps_2_2 is compatible with Deps_1, which is an example of following the Yes branch of step **318** in FIG. **3**B. The checkmark further indicates that Deps_2_2 will be added to an extended range of versions to which Deps_1 is mapped. The "x" in the box in FIG. **5**C indicates that running the validation script associated with Library_B determines that Deps_2_2 is not compatible with Library_B, which is an example of following the No branch of step **318** in FIG. **3**B. The "x" in the box further indicates that there will be no change to the range of versions to which Library_B is mapped.

**[0071]** In FIG. **5**D, example **500** includes a step **3**, which includes validation component **506** determining whether the validation of Deps_1 is successful and if the validation is successful, step **3** further includes dependency version tree update module **208** updating the dependency version tree to include Deps_2_2. Step **3** further includes validation component **506** recursively proceeding to verify the compatibility of other libraries according to a direction toward the root node of the dependency version tree. More specifically, step **3** includes validation component **506** determining that the validation of Deps_1 is successful, and dependency version tree update module **208** updating the dependency version tree to include Deps_2_2 in an extended range of versions to which Deps_1 is mapped. This extended range of versions includes both Deps_2_1 and Deps_2_2, and is illustrated in the mapping of Deps_1 in mapping storage **508** in FIG. **5**D. Step **3** is an example of steps **318** and **320** in FIG. **3**B.

**[0072]** Following the update of the dependency version tree, step **3** in FIG. **5**D also includes validation component **506** identifying a set of libraries **542**, which are all the libraries that depend on Deps_1 (i.e., identifying Library_A and Library_C) and applying the validation process to all the identified libraries that depend on Deps_1, where the validation process is the same as the validation process that was applied to Deps_1. The validation process applied to Library_A includes verifying compatibility between Deps_2_2 and Library_A and results in a successful validation, as indicated by the checkmark. The validation process applied to Library_C includes verifying compatibility between Deps_2_2 and Library_C and results in an unsuccessful validation, as indicated by the "x" in the box.

**[0073]** In FIG. **5**E, Library_A in set of libraries **542** that depend on Deps_1 is mapped to an extended range of versions that includes both Deps_2_1 and Deps_2_2, which is a result of the successful validation discussed above relative to FIG. **5**D. Library_C in set of libraries **542** retains an original mapping to Deps_2_1 and does not have a mapping to Deps_2_2, which is a result of the unsuccessful validation discussed above relative to FIG. **5**D. FIG. **5**E illustrates a result of step **322** in FIG. **3**B.

**[0074]** FIG. **6** is an example **600** of queueing different updates of library dependencies based on respective positions in the dependency version tree, in accordance with embodiments of the present invention. Example **600** includes a developer **602** of Library_A, a developer **604** of Deps_1, a public library repository **606**, a validation component **608**, and a mapping storage **610**. Public library repository **606** is an example of public library repository **404**, validation component **608** is an example of validation component **206**, and mapping storage **610** is an example of mapping storage **406**. Mapping storage includes a mapping of library **612** (i.e., Library_A) to Deps_1 and Deps_2_1 and a mapping of library **614** (i.e., Deps_1) to Deps_2_1, Deps_3, and Deps_4.

**[0075]** Example **600** illustrates a case in which developer **602** of Library_A and developer **604** of Deps_1 submit new library versions at the same time or nearly at the same time. That is, developer **602** submits an upgrade of Deps_1 and developer **604** submits an upgrade of **604**. Because the dependency version tree needs to be made consistent, parallel processing of the two upgrades is not allowed. Instead, a prioritization policy is employed to queue the upgrades in a particular sequence. In this case, the processing of the upgrade of the dependency of Library_A is queued to wait behind the processing of the upgrade of the dependency of Deps_1. Thus, the processing for Library_A begins only after the validation for Deps_1 has been completed in steps **316**, **318**, **320**, and **322** in FIG. **3**A and FIG. **3**B. The prioritization policy specifies that when updating dependencies of given libraries, processing a first given library at a leaf node or node further from a root node of the dependency version tree is given priority over processing a second given library at a root node or node closer to a root node of the dependency version tree.

**[0076]** In one embodiment, an employment of the prioritization policy is added to the process of FIGS. **3**A-**3**B by including the following operations: (i) receiving multiple requests for upgrading respective libraries in the public library repository; (ii) determining respective nodes in the dependency version tree that are associated with the multiple requests; (iii) determining a sequence of the respective nodes based on distances between each of the respective nodes and a root node of the dependency version tree; (iv) determining a queue of the multiple requests according to the sequence based on the distances, so that a first request is positioned earlier in the queue than a second request, based on a first node associated with the first request being further from a root node in the dependency version tree than a second node associated with the second request; and (v) validating the respective libraries in an order specified by the queue, so that a validation of one library associated with one request in the queue of the multiple requests is completed before a beginning of a processing and a validation of a next library associated with a next request in the queue of the multiple requests.

**[0077]** FIG. **7** is an example **700** of installing a library and library dependencies in the process of FIG. **3**C, in accordance with embodiments of the present invention. Example **700** includes a local computer system **702**, a public library repository **704**, a mapping storage **706**, and a user **708** that utilizes computer system **702**. Public library repository **704** is an example of public library repository **404** and mapping storage **706** is an example of mapping storage **406**. Example **700** includes a step **1** that includes user **708** using computer system **702** to request an installation of Library_A locally at computer system **702**. Step **1** is an example of an action included in step **352** in FIG. **3**C.

**[0078]** Example **700** further includes a step **2** that includes computer system **702** downloading Library_A and down-

loading the dependency version tree from public library repository **704**. Step 2 is an example of an action included in step **354** in FIG. **3**C.

**[0079]** Example **700** further includes a step **3** that includes computer system **702** installing specific dependent libraries (i.e., libraries on which Library_A depends) according to the dependency version tree, and then installing Library_A. Step **3** is an example of an action included in step **356** in FIG. **3**C. Although not shown in FIG. **7**, following step **3**, user **708** uses the newly installed Library_A via computer **702** as part of a software development activity. Based on the utilization of the dependency version tree, user **708** ensures that there are no conflicts between dependencies of Library_A during the installation of Library_A in step **3** and while using Library_A. The use of the installed Library_A is an example of an action included in step **358** in FIG. **3**C.

**[0080]** The library development technique disclosed herein allows users to specify a library version selection policy. In one embodiment, the library version selection policy includes the options "latest version compatible," "oldest version compatible," "median version compatible," and "random version compatible." These options correspond to selecting the latest compatible version of the library, the oldest compatible version of the library, a version in the middle range of compatibility, and a randomly selected compatible version of the library, respectively. After the specification of the library version selection policy, computer system **702** uses the user-specified library version selection policy to determine and provide an available dependency version tree. Because the library version selection policy is applied automatically via the dependency version tree and because the download of the dependency version tree is not visible to users, users do not need to consider the issue of multiple versions being available for a given library (i.e., a user does not need to use conventional techniques that include the user considering documentation of the ranges of versions of libraries).

**[0081]** The descriptions of the various embodiments of the present invention have been presented herein for purposes of illustration but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those or ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

What is claimed is:

**1**. A computer-implemented method comprising:

receiving, by a processor set, requests to publish multiple libraries to a public library repository, the requests including specifications of respective sets of one or more dependencies of the multiple libraries and respective validation scripts;

building, by the processor set, a dependency version tree that includes the specifications of the respective sets of one or more dependencies;

receiving, by the processor set, a request to upload a new version of a library to the public library repository, the library being included in the multiple libraries; and

validating, by a validation component and using the dependency version tree and the validation scripts, a compatibility between the library and one or more libraries dependent upon the library.

**2**. The method of claim **1**, wherein the validating includes:

identifying the one or more libraries dependent upon the library by using the dependency version tree; and

verifying the compatibility between the library and the one or more libraries dependent upon the library by invoking one or more validation scripts associated with the one or more libraries, the one or more validation scripts being included in the respective validation scripts, wherein the invoking the one or more validation scripts includes verifying whether or not the library is compatible with the one or more libraries.

**3**. The method of claim **1**, further comprising:

receiving, by the processor set, a request to install another library, the other library being dependent upon the new version of the library;

downloading, by the processor set, the other library and the dependency version tree;

installing, by the processor set and based on the downloaded dependency version tree, one or more dependent libraries that are dependent upon the other library and subsequently installing, by the processor set, the other library; and

using the installed other library with an assurance of a compatibility between the other library and the one or more dependent libraries and with a minimized likelihood of a conflict between the other library and any one of the one or more dependent libraries.

**4**. The method of claim **1**, further comprising:

determining, by the processor set, that the validating by the validation component indicates a successful validation of the compatibility between the library and another library included in the one or more libraries; and

based on the successful validation of the compatibility, updating, by the processor set, the dependency version tree to include the new version in an extended range of versions to which the other library is mapped.

**5**. The method of claim **4**, further comprising:

recursively validating, by the validation component, a compatibility of the other library with one or more other libraries dependent upon the other library, wherein the recursively validating includes traversing a path in the dependency version tree by moving from a node of the dependency version tree indicating the other library to a root node of the dependency version tree.

**6**. The method of claim **1**, further comprising:

receiving, by the processor set, an upload of the new version of the library to the public library repository; and

detecting, by the validation component, the upload of the new version, wherein the validating the compatibility between the library and the one or more libraries is performed in response to the detecting.

**7**. The method of claim **1**, further comprising:

receiving, by the processor set, multiple requests for upgrading respective libraries in the public library repository;

determining, by the processor set, respective nodes in the dependency version tree that are associated with the requests;

determining a sequence of the respective nodes based on distances between each of the respective nodes and a root node of the dependency version tree;

determining a queue of the multiple requests according to the sequence based on the distances; and

validating the respective libraries in an order specified by the queue, so that a validation of one library associated with one request in the queue of the multiple requests is completed before a beginning of a processing and a validation of a next library associated with a next request in the queue of the multiple requests.

**8**. A computer system comprising:

a processor set;

a set of one or more computer-readable storage media; and

program instructions, collectively stored in the set of one or more computer-readable storage media, for causing the processor set to perform the following computer operations:

receive requests to publish multiple libraries to a public library repository, the requests including specifications of respective sets of one or more dependencies of the multiple libraries and respective validation scripts;

build a dependency version tree that includes the specifications of the respective sets of one or more dependencies;

receive a request to upload a new version of a library to the public library repository, the library being included in the multiple libraries; and

validate, by using the dependency version tree and the validation scripts, a compatibility between the library and one or more libraries dependent upon the library.

**9**. The computer system of claim **8**, wherein the computer operation of validate the dependency version tree includes the following additional computer operations:

identify the one or more libraries dependent upon the library by using the dependency version tree; and

verify the compatibility between the library and the one or more libraries dependent upon the library by invoking one or more validation scripts associated with the one or more libraries, the one or more validation scripts being included in the respective validation scripts, wherein the invoking the one or more validation scripts includes verifying whether or not the library is compatible with the one or more libraries.

**10**. The computer system of claim **8**, wherein the program instructions cause the processor set to perform the following additional computer operations:

receive a request to install another library, the other library being dependent upon the new version of the library;

download the other library and the dependency version tree;

install, based on the downloaded dependency version tree, one or more dependent libraries that are dependent upon the other library and subsequently install the other library; and

use the installed other library with an assurance of a compatibility between the other library and the one or more dependent libraries and with a minimized likelihood of a conflict between the other library and any one of the one or more dependent libraries.

**11**. The computer system of claim **8**, wherein the program instructions cause the processor set to perform the following additional computer operations:

determine that a performance of the computer operation of validate the compatibility indicates a successful validation of the compatibility between the library and another library included in the one or more libraries; and

based on the successful validation of the compatibility, update the dependency version tree to include the new version in an extended range of versions to which the other library is mapped.

**12**. The computer system of claim **11**, wherein the program instructions cause the processor set to perform the following additional computer operations:

recursively validate, using the validation component, a compatibility of the other library with one or more other libraries dependent upon the other library, wherein the computer operation of recursively validate the compatibility includes the computer operation of traverse a path in the dependency version tree by moving from a node of the dependency version tree indicating the other library to a root node of the dependency version tree.

**13**. The computer system of claim **8**, wherein the program instructions cause the processor set to perform the following additional computer operations:

receive an upload of the new version of the library to the public library repository; and

detect the upload of the new version, wherein the computer operation of validate the compatibility between the library and the one or more libraries is performed in response to detecting the upload of the new version.

**14**. The computer system of claim **8**, wherein the program instructions cause the processor set to perform the following additional computer operations:

receive multiple requests for upgrading respective libraries in the public library repository;

determine respective nodes in the dependency version tree that are associated with the requests;

determine a sequence of the respective nodes based on distances between each of the respective nodes and a root node of the dependency version tree;

determine a queue of the multiple requests according to the sequence based on the distances; and

validate the respective libraries in an order specified by the queue, so that a validation of one library associated with one request in the queue of the multiple requests is completed before a beginning of a processing and a validation of a next library associated with a next request in the queue of the multiple requests.

**15**. A computer program product comprising:

a set of one or more computer-readable storage media; and

program instructions, collectively stored in the set of one or more computer-readable storage media, for causing a processor set to perform the following computer operations:

receive requests to publish multiple libraries to a public library repository, the requests including specifications of respective sets of one or more dependencies of the multiple libraries and respective validation scripts;

build a dependency version tree that includes the specifications of the respective sets of one or more dependencies;

receive a request to upload a new version of a library to the public library repository, the library being included in the multiple libraries; and

validate, by using a validation component, the dependency version tree, and the validation scripts, a compatibility between the library and one or more libraries dependent upon the library.

**16**. The computer program product of claim **15**, wherein the computer operation of validate the dependency version tree includes the following additional computer operations:

identify the one or more libraries dependent upon the library by using the dependency version tree; and

verify the compatibility between the library and the one or more libraries dependent upon the library by invoking one or more validation scripts associated with the one or more libraries, the one or more validation scripts being included in the respective validation scripts, wherein the invoking the one or more validation scripts includes verifying whether or not the library is compatible with the one or more libraries.

**17**. The computer program product of claim **15**, wherein the program instructions cause the processor set to perform the following additional computer operations:

receive a request to install another library, the other library being dependent upon the new version of the library;

download the other library and the dependency version tree;

install, based on the downloaded dependency version tree, one or more dependent libraries that are dependent upon the other library and subsequently install the other library; and

use the installed other library with an assurance of a compatibility between the other library and the one or more dependent libraries and with a minimized likelihood of a conflict between the other library and any one of the one or more dependent libraries.

**18**. The computer program product of claim **15**, wherein the program instructions cause the processor set to perform the following additional computer operations:

determine that a performance of the computer operation of validate the compatibility indicates a successful validation of the compatibility between the library and another library included in the one or more libraries; and

based on the successful validation of the compatibility, update the dependency version tree to include the new version in an extended range of versions to which the other library is mapped.

**19**. The computer program product of claim **18**, wherein the program instructions cause the processor set to perform the following additional computer operations:

recursively validate, using the validation component, a compatibility of the other library with one or more other libraries dependent upon the other library, wherein the computer operation of recursively validate the compatibility includes the computer operation of traverse a path in the dependency version tree by moving from a node of the dependency version tree indicating the other library to a root node of the dependency version tree.

**20**. The computer program product of claim **15**, wherein the program instructions cause the processor set to perform the following additional computer operations:

receive an upload of the new version of the library to the public library repository; and

detect the upload of the new version, wherein the computer operation of validate the compatibility between the library and the one or more libraries is performed in response to detecting the upload of the new version.

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